The regularity method and blow-up lemmas for sparse graphs

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Course outline

1. **Lecture I.** Introduction to the regularity method
2. **Lecture II.** The blow-up lemma
3. **Lecture III.** The sparse case: small subgraphs
4. **Lecture IV.** The sparse case: large subgraphs

Lecture I. Introduction to the regularity method

Introduction

Regularity method:

- A powerful method in graph theory and combinatorics and beyond.
- Shall focus on graphs, including some applications in the theory of random and pseudorandom graphs (Lecture III).
- Shall consider blow-up lemmas in the dense and sparse settings (Lectures II & IV).
- Lecture I: introduction to the regularity method

Outline (Lecture I)

1. Basic definitions
2. The regularity lemma
3. Embedding/counting lemmas
4. Example applications
5. Some words on proofs

Basic definitions

Definition 1 (Basic definition)**.** $G = (V, E)$ *a graph;* $U, W \subset V$ *non-empty and disjoint. Say* (U, W) *is* ε*-regular* (*in* G) *if*

- *for all* $U' \subset U$, $W' \subset W$ *with* $|U'| \geq \varepsilon|U|$ *and* $|W'| \geq \varepsilon|W|$, *we have*

$$\left| \frac{|E(U', W')|}{|U'||W'|} - \frac{|E(U, W)|}{|U||W|} \right| \leq \varepsilon.$$

Density:

$$d(U, W) = \frac{e(U, W)}{|U||W|} = \frac{|E(U, W)|}{|U||W|}$$

Basic definitions

Definition 2. *A partition* $V = V_1 \cup \cdots \cup V_t$ *is an equipartition if*

$$|V_1| \leq \cdots \leq |V_t| \leq |V_1| + 1.$$

Definition 3. *A partition* $V = V_1 \cup \cdots \cup V_t$ *is* ε*-regular if at least* $(1-\varepsilon)\binom{t}{2}$ *pairs* (V_i, V_j) $(i < j)$ *are* ε*-regular.*

Szemerédi's regularity lemma

Theorem 4 (The regularity lemma)**.** *For any $\varepsilon > 0$ and $t_0 \geq 1$, there exist T_0 such that any graph G admits an ε-regular equipartition $V = V_1 \cup \cdots \cup V_t$ with $t_0 \leq t \leq T_0$.*

Crucial: T_0 is independent of $|V(G)|$.

Earlier version: important lemma in Szemerédi's proof of the Erdős–Turán conjecture on arithmetic progressions in sets of integers of positive density.

Komlós:

> 'This is not a very transparent theorem, but it grows on you with time.'

Exceptional class

Sometimes consider partitions as follows:

(*i*) $V = V_0 \cup V_1 \cup \cdots \cup V_t$,

(*ii*) $|V_0| \leq \varepsilon |V|$,

(*iii*) $|V_1| = \cdots = |V_t|$.

Can even demand $|V_0| < t$. Regularity lemma delivers such partitions, as long as $|V| \geq n_0(t_0, \varepsilon)$.

Exceptional class: V_0

The embedding lemma/counting lemma (simplest version)

Set-up 5. *Let* $G = (V, E)$ *be a graph with*

(*i*) $V = V_1 \cup \cdots \cup V_r$ *and* $|V_i| = m$ *for all* i,

(*ii*) (V_i, V_j) ε*-regular for all* $i < j$,

(*iii*) $|E(V_i, V_j)| = \rho m^2$ *for all* $i < j$.

Write $G_r^{(\varepsilon)}(m, \rho)$ for a graph as above.

The embedding lemma/counting lemma (simplest version)

Let K^r have vertices $x_1, \dots, x_r$.

Lemma 6 (Embedding lemma/Counting lemma)**.** $\forall r \geq 2$, $\rho > 0$ *and* $\delta > 0$ $\exists \varepsilon > 0$, m_0: *if* $m \geq m_0$, *then the number of homomorphisms* $K^r \hookrightarrow G = G_r^{(\varepsilon)}(m, \rho)$ *with* $x_i \mapsto$ *some* $v_i \in V_i$ *for all* i *is*

$$m^r \rho^{\binom{r}{2}} \pm \delta m^r.$$

In particular, $K^r \subset G$.

The reduced graph R

After regularization, can consider the so-called *reduced graph*.

- ▷ Vertex i for each V_i ($1 \leq i \leq t$),
- ▷ Edge-weight $d(V_i, V_j)$ for each $1 \leq i < j \leq t$ that is ε-regular and *dense* (usually, use some cut-off density as threshold).

We can use R to define a 'generalized random graph' G^{rand} on $V(G)$: replace each edge ij of R by a random bipartite graph on $V_i \times V_j$ with edge probability $d(V_i, V_j)$.

G and G^{rand} are 'similar'

- For any *fixed* graph H, the number of copies of H in G and in G^{rand} are approximately equal, as long as $n = |V(G)|$ is large.
- For any $\delta > 0$ and h, there are ε and ε' such that if R is obtained from SzRL with parameter ε and cut-off ε', then, for any $H = H^h$,

$$\#\{H \subset G\} = \mathbb{E}(\#\{H \subset G^{rand}\}) \pm \delta n^h.$$

- Follows from 'counting lemmas' **[exercise!]**.
- How about *large* H? E.g., $h = n$? Lectures II & IV will focus on 'embedding lemmas' for such H ('blow-up lemmas').

The embedding lemma/counting lemma (exercises)

Exercise 7 (Counting lemma)**.** *What is the CL for a general graph* $H = H^r$*? Suppose* H *has vertices* $x_1, \dots, x_r$*. What is the number of homomorphisms* $H \hookrightarrow G_r^{(\varepsilon)}(m, \rho)$ *with* $x_i \mapsto$ *some* $v_i \in V_i$ *for all* i*?*

Exercise 8 (Counting lemma)**.** *What is the CL for a general* r*-partite graph* $H = H^\ell$*? Suppose* H *has vertices* $x_1, \dots, x_\ell$*. Suppose* $\chi : V(H) \to [r] = \{1, \dots, r\}$ *is an* r*-partition of* H*. What is the number of homomorphisms* $H \hookrightarrow G_r^{(\varepsilon)}(m, \rho)$ *with* $x_i \mapsto$ *some* $v_i \in V_{\chi(i)}$ *for all* i*?*

Could also consider $G_r^{(\varepsilon)}(m, (\rho_{ij})_{1 \le i < j \le r})$.

The embedding lemma/counting lemma (exercises)

Exercise 9 (Counting lemma (induced version)). *What is the induced CL for a general graph* $H = H^r$*? Suppose* H *has vertices* $x_1, \dots, x_r$*. What is the number of homomorphisms* $H \hookrightarrow G_r^{(\varepsilon)}(m, \rho)$ *with* $x_i \mapsto$ *some* $v_i \in V_i$ *for all* i *such that* H *is isomorphic to* $G[v_1, \dots, v_r]$*?*

Could also consider r-partite graphs $H = H^\ell$.

A few applications of the regularity method

(*i*) The removal lemma

(*ii*) The Erdős–Stone–Simonovits theorem

Triangle removal lemma

Theorem 10 (Triangle removal lemma)**.** $\forall \varepsilon > 0\ \exists \delta > 0$ *for which the following holds. Suppose* $G = G^n = (V, E)$ *is such that* $G - F = (V, E \setminus F)$ *contains a triangle* $\forall F \subset \binom{V}{2}$ *with* $|F| \leq \varepsilon \binom{n}{2}$*. Then* G *contains* $\geq \delta n^3$ *triangles.*

That is: $G = G^n$ ε-far from K^3-free, then $\#\{K^3 \hookrightarrow G\} \geq \delta n^3$.

Exercise 11. *Deduce the triangle removal lemma from the regularity lemma.*

Hint 12. *Analyse the 'cleaned-up graph'* G^* (*Definition 19*)*.*

Exercise 13. *State and prove the removal lemma for general graphs* H*.*

Triangle removal lemma

Exercise 14 (Roth's theorem)**.** $\forall \delta > 0\ \exists n_0$ *such that, for all* $n \geq n_0$*, if* $A \subset [n]$ *and* $|A| \geq \delta n$*, then* A *contains a* 3*-term arithmetic progression* $\{a, a+d, a+2d\}\ (d > 0)$*.*

Hint 15. *Consider a* 3*-partite graph* $G = (X, Y, Z; E)$ *defined as follows. Let* $X = [n]$*,* $Y = [2n]$ *and* $Z = [3n]$ *(disjoint). For each* $a \in A$*, put in* G *the edges* $(x, x+a) \in X \times Y$*,* $(x, x+2a) \in X \times Z$ *and* $(y, y+a) \in Y \times Z$*. What happens if* G *contains a triangle* ***not*** *of the form* $\{x, x+a, x+2a\}$*?*

The Erdős–Stone–Simonovits theorem

- Let $\mathrm{ex}(n, F) = \max\{e(G) \colon F \not\subset G \subset K^n\}$. Mantel: $\mathrm{ex}(n, K^3) = \lfloor n/2 \rfloor \lceil n/2 \rceil$.
- Mantel implies that $1/2$ is the *density threshold* for K^3: if G has "density" $1/2 + \varepsilon$, then $G \supset K^3$ (if n large). Indeed,

$$\mathrm{ex}(n, K^3) = \left(\frac{1}{2} + o(1)\right)\binom{n}{2}.$$

- Turán: what is $\mathrm{ex}(n, K^{q+1})$? Turán graph T_q^n: split n vertices into q classes as equally as possible; add all edges connecting vertices in different classes. Lower bound:

$$\mathrm{ex}(n, K^{q+1}) \geq e(T_q^n).$$

The Erdős–Stone–Simonovits theorem

Theorem 16 (Turán (1941)). *For all* n *and* q,

$$ex(n, K^{q+1}) = e(T_q^n).$$

Density threshold for K^{q+1} is $1 - 1/q$.

How about arbitrary F?

The Erdős–Stone–Simonovits theorem

F: an arbitrary graph ($e(F) \geq 1$). Lower bound for $\mathrm{ex}(n, F)$?

$\chi(F)$: the *chromatic number* of F, that is, the smallest q such that $F \subset T_q^\infty$

Lower bound: $\mathrm{ex}(n, F) \geq e\left(T^n_{\chi(F)-1}\right) = \left(1 - \frac{1}{\chi(F)-1} + o(1)\right)\binom{n}{2}$

Theorem 17 (Erdős & Stone 1946, Erdős and Simonovits 1966)**.** *For every graph F, we have*

$$\mathrm{ex}(n, F) = \left(1 - \frac{1}{\chi(F) - 1} + o(1)\right)\binom{n}{2}.$$

Density threshold for F is $1 - \frac{1}{\chi(F)-1}$.

The Erdős–Stone–Simonovits theorem

Exercise 18. *Deduce the Erdős–Stone–Simonovits theorem from Turán's theorem and the regularity lemma.*

Outline: $G = G^n$ with $e(G) \geq (1 - 1/(q-1) + \eta)\binom{n}{2}$, where $q = \chi(G)$.

(*i*) *Regularize* G: apply Szemerédi's regularity lemma to G (use $\varepsilon \ll \eta$)

(*ii*) Analyse the 'cleaned-up graph' G^* (Definition 19). Deduce from Turán's theorem that $G_q^{(\varepsilon)}(m, \rho) \subset G$.

(*iii*) Apply the counting lemma.

Terminology: *Cleaned-up graph* G^*

Definition 19. *After regularization of* G*, have* $V = V_1 \cup \cdots \cup V_t$*. Remove all edges in* $G[V_i, V_j]$ *for all* i *and* j *such that*

1. (V_i, V_j) *is not* ε*-regular,*
2. $|E(V_i, V_j)| \leq f(\varepsilon)|V_i||V_j|$ (*suitable* f *with* $f(\varepsilon) \to 0$ *as* $\varepsilon \to 0$).

Resulting graph: cleaned-up graph G^**.*

In G^*, every $G^*[V_i, V_j]$ is regular and 'dense'. Usually, lose very little.

A Ramsey–Turán result

Exercise 20 (Szemerédi 1972)**.** *Suppose* $G = G^n$ *has independence number* $o(n)$ *and* $e(G) \geq (1/8+\varepsilon)n^2$ *for some fixed* $\varepsilon > 0$*. Prove that* G *contains a* K^4*.*

Further exercises

Notation/definitions:

(*i*) $G \to H$: if $E(G) = B \cup R$, then $H \subset G[B]$ or $H \subset G[R]$.

(*ii*) $G \xrightarrow{\text{ind}} H$: if $E(G) = B \cup R$, then there is an induced subgraph H' of G isomorphic to H with $H' \subset G[B]$ or $H' \subset G[R]$.

(*iii*) The (2-*colour*) *Ramsey number* of H is

$$r(H) = \min\{|V(G)|\colon G \to H\} = \min\{n\colon K^n \to H\}.$$

Exercise 21. *Prove that, for every* H*, there is* G *such that* $G \xrightarrow{\text{ind}} H$.

Another application

Theorem 22 (Chvátal, Rödl, Szemerédi and Trotter 1983)**.** *For every* Δ*, there is* $c = c_\Delta$ *such that, for every graph* $H = H^\ell$ *with* $\Delta(H) \le \Delta$*, we have*

$$r(H) \le c\ell.$$

That is, bounded degree graphs have linear Ramsey numbers.

Recall

$$2^{\ell/2} \le r(K^\ell) \le 4^\ell.$$

Theorem 22: typical application of the regularity method.

Positive proportion EL/CL

Important: suppose x_i to be embedded in G, and $Y \subset N_H(x_i)$ already embedded. Then have the set of *candidate vertices*

$$C(x_i) = V_{\chi(i)} \cap \bigcap_{y \in Y} N_G(y).$$

More precisely, have the set of *available vertices*

$$A(x_i) = V_{\chi(i)} \cap \bigcap_{y \in Y} N_G(y) \setminus \operatorname{Im} \psi,$$

where ψ is the current partial embedding. Can make the sets $C(x_i)$ shrink in a controlled way: $|C(x_i)| \approx \rho^{|Y|}|V_i|$. When $H = H^\ell$ and $\ell = O(1)$, then $A(x_i) \approx C(x_i)$. How about if $\ell \to \infty$?

Positive proportion EL/CL

Natural restriction. Suppose $H = H^\ell$ has bounded maximum degree: $\Delta(H) = O(1)$ (i.e., $\Delta = \Delta(H)$ independent of $n = |V(G)| \to \infty$), but allow $\ell \to \infty$.

Let $H = H^\ell$ be r-partite and suppose $V(H) = \{x_1, \ldots, x_\ell\}$. Suppose

$$\chi\colon V(H) \to [r] = \{1, \ldots, r\}$$

is an r-partition of H. We think of r as fixed and $\ell \to \infty$.

Positive proportion EL/CL

Lemma 23 (Embedding lemma/Counting lemma)**.** $\forall r \geq 2$, $\ell \geq 1$, $\rho > 0$, $\delta > 0$ *and* Δ $\exists \varepsilon > 0$, m_0: *if* $\Delta(H) \leq \Delta$ *and* $m \geq m_0$, *then the number of homomorphisms* $H \hookrightarrow G = G_k^{(\varepsilon)}(m, \rho)$ *with* $x_i \mapsto$ *some* $v_i \in V_{\chi(i)}$ *for all* i *is*

$$m^\ell \rho^{e(H)} \pm \delta m^\ell.$$

In particular, $H \subset G$.

Suffices: $m_0 \gg \ell \rho^{-\Delta}$

Exercise 24. *Deduce the Chvátal–Rödl–Szemerédi–Trotter theorem from Lemma 23.*

Almost spanning embedding lemma

Challenge 25. *State and prove an almost spanning embedding lemma: a version of the embedding lemma in which the vertex classes* $X_1, \dots, X_r$ *of* H *are each of size* $.99m$*. Here, we suppose* $r = O(1)$ *and* $\Delta(H) = O(1)$*.*

Proof of the regularity lemma (rough sketch)

Let $G = (V, E)$ and $\varepsilon > 0$ be fixed. Let $P = (C_i)_{0 \le i \le k}$ be an equitable partition of V ($V = C_0 \cup \cdots \cup C_k$). For each ε-irregular pair (C_s, C_t) with $1 \le s < t \le k$, choose $X(s,t) \subset C_s$, $Y(s,t) \subset C_t$ witnessing this fact.

For fixed $1 \le s \le k$, the sets $X(s,t)$ in

$$\{X(s,t) \subset C_s : 1 \le t \le k \text{ and } (C_s, C_t) \text{ is not } \varepsilon\text{-regular}\}$$

define a natural partition of C_s into at most 2^{k-1} blocks, called *atoms*. Put all of these atoms together to form an equitable partition $Q = (C'_i)_{0 \le i \le k'}$, with $k' = k4^k$ and $|C'_0| \le |C_0| + n4^{-k}$.

Proof of the regularity lemma (rough sketch)

Definition 26. *The index* $\mathrm{ind}(R)$ *of an equitable partition* $R = (C_i)_0^r$ *of* V *is*

$$\mathrm{ind}(R) = \frac{2}{r^2} \sum_{1 \le i < j \le \ell} d(C_i, C_j)^2.$$

Trivially, $0 \le \mathrm{ind}(R) < 1$.

Proof of the regularity lemma (rough sketch)

Recall we constructed an equipartition $Q = Q(P)$ from P.

Lemma 27. *If P is not ε-regular, then*

$$\mathrm{ind}(Q) \geq \mathrm{ind}(P) + 10^{-2}\varepsilon^5.$$

Conclusion: we can't iterate more than $10^2/\varepsilon^5$ times!

Proof of the regularity lemma (rough sketch)

Where does the gain come from?

Lemma 28. *Let* $y_1, \dots, y_v \geq 0$ *be given. Suppose* $0 \leq \rho = u/v < 1$, *and* $\sum_{1 \leq i \leq u} y_i = \alpha\rho \sum_{1 \leq i \leq v} y_i$. *Then*

$$\sum_{1 \leq i \leq v} y_i^2 \geq \frac{1}{v}\left(1 + (\alpha - 1)^2 \frac{\rho}{1-\rho}\right)\left\{\sum_{1 \leq i \leq v} y_i\right\}^2.$$

Each irregular pair contributes $\approx \varepsilon^4$ to the index. An ε-fraction of the pairs contributes that much; total is ε^5, as claimed.

Summary

- ▷ The regularity method
- ▷ The regularity lemma + embedding lemmas
- ▷ Example applications

Lecture II:

- ○ The blow-up lemma: embedding large graphs

The regularity method and blow-up lemmas for sparse graphs

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Lecture II. The blow-up lemma

From Lecture I

- Elements of the regularity method
- The regularity lemma

$$\forall G : G \approx G^{rand} \text{ (w.r.t. small subgraph statistics)}$$

- Embedding lemmas

$$\forall \text{ bdd deg \& sufficiently small } H \ \& \ G_r^{(\varepsilon)}(m, \rho) : H \hookrightarrow G_r^{(\varepsilon)}(m, \rho)$$

Applications of the the blow-up lemma

So far: we saw the regularity method applied to problems involving subgraphs $H = H^{\ell}$ of some $G = G^{n}$ with $\ell = o(n)$.

We are now concerned with the case in which $\ell \approx n$ or even $\ell = n$ (almost spanning or spanning subgraph problems).

(*i*) The Alon–Yuster conjecture

(*ii*) The Pósa–Seymour conjecture

The Alon–Yuster theorem & conjecture

Theorem 1 (Alon and Yuster (1997)). $\forall \varepsilon > 0,\ h \geq 1\ \exists n_0 = n_0(\varepsilon, h)$ *such that, for every* $H = H^h$ *and* $n \geq n_0$, *any* $G = G^{hn}$ *with minimum degree* $\delta(G) \geq (1 - 1/\chi(H) + \varepsilon)hn$ *has an* H*-factor.*

Conjecture 2. $\delta(G) \geq (1 - 1/\chi(H))nh + c(h)$ *suffices.*

The Pósa–Seymour conjecture

Conjecture 3. *Suppose* $G = G^n$. *If* $\delta(G) \geq (1-1/(k+1))n$, *then* G *contains the* k*th power of a Hamilton cycle.*

Theorem 4 (Hajnal–Szemerédi (1970))**.** *Suppose* $G = G^n = G^{s(k+1)}$. *If* $\delta(G) \geq (1-1/(k+1))n$, *then* $G \supset sK^{k+1}$.

Theorem 5 (Corrádi–Hajnal (1963))**.** *Suppose* $G = G^{3s}$. *If* $\delta(G) \geq 2s$, *then* $G \supset sK^3$.

Komlós–Sárközy–Szemerédi theorems

Theorem 6. *The Alon–Yuster conjecture holds for all* $n \geq n_0(H)$.

Theorem 7. *The Pósa–Seymour conjecture holds for all* $n \geq n_0(k)$.

A key tool: the blow-up lemma

Before we proceed:

Exercise 8. *Prove that the minimum degree conditions are best possible.*

Basic definition

Definition 9 ((ε,δ)-super-regular)**.** *Fix ε and $\delta > 0$. The bipartite graph $G = (V_1, V_2; E)$ with $|V_1| = |V_2| = n$ is an (ε,δ)-super-regular pair if*

- ▷ *for all $W_1 \subset V_1$, $W_2 \subset V_2$ with $|W_1|, |W_2| \geq \varepsilon n$, we have*

$$|d(W_1, W_2) - d(V_1, V_2)| \leq \varepsilon,$$

- ▷ $\deg(v) \geq \delta n$ *for all $v \in V_1 \cup V_2$.*

Terminology:

- ◦ ‘regular’: edges uniformly distributed
- ◦ ‘super’: high minimum degree

Exercises

Exercise 10. *Prove that, for any positive* ρ *and* δ*, there are* $\varepsilon > 0$ *and* n_0 *for which the following holds. Suppose* $G = (U, W; E)$ *is an* (ε, δ)*-super-regular pair with* $|U| = |W| \geq n_0$ *and density* $d(U, W) = e(U, W)/|U||W| \geq \rho$*. Then* G *has a perfect matching.*

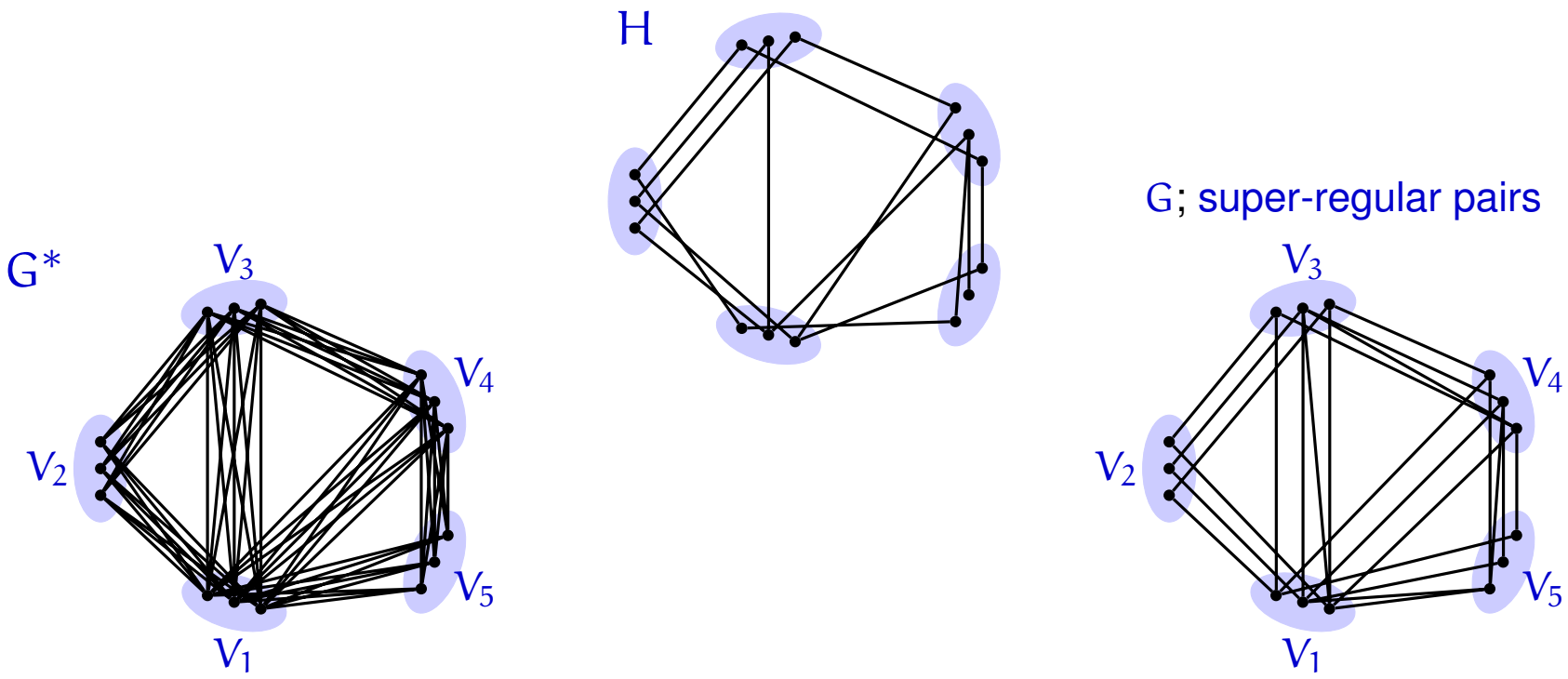
You can also try your teeth on the following:

Exercise 11. *Prove that, for any positive* ρ *and* δ*, there are* $\varepsilon > 0$ *and* n_0 *for which the following holds. Suppose* $G = (U, W; E)$ *is an* (ε, δ)*-super-regular pair with* $|U| = |W| \geq n_0$ *and density* $d(U, W) = e(U, W)/|U||W| \geq \rho$*. Then* G *has a Hamilton path/cycle.*

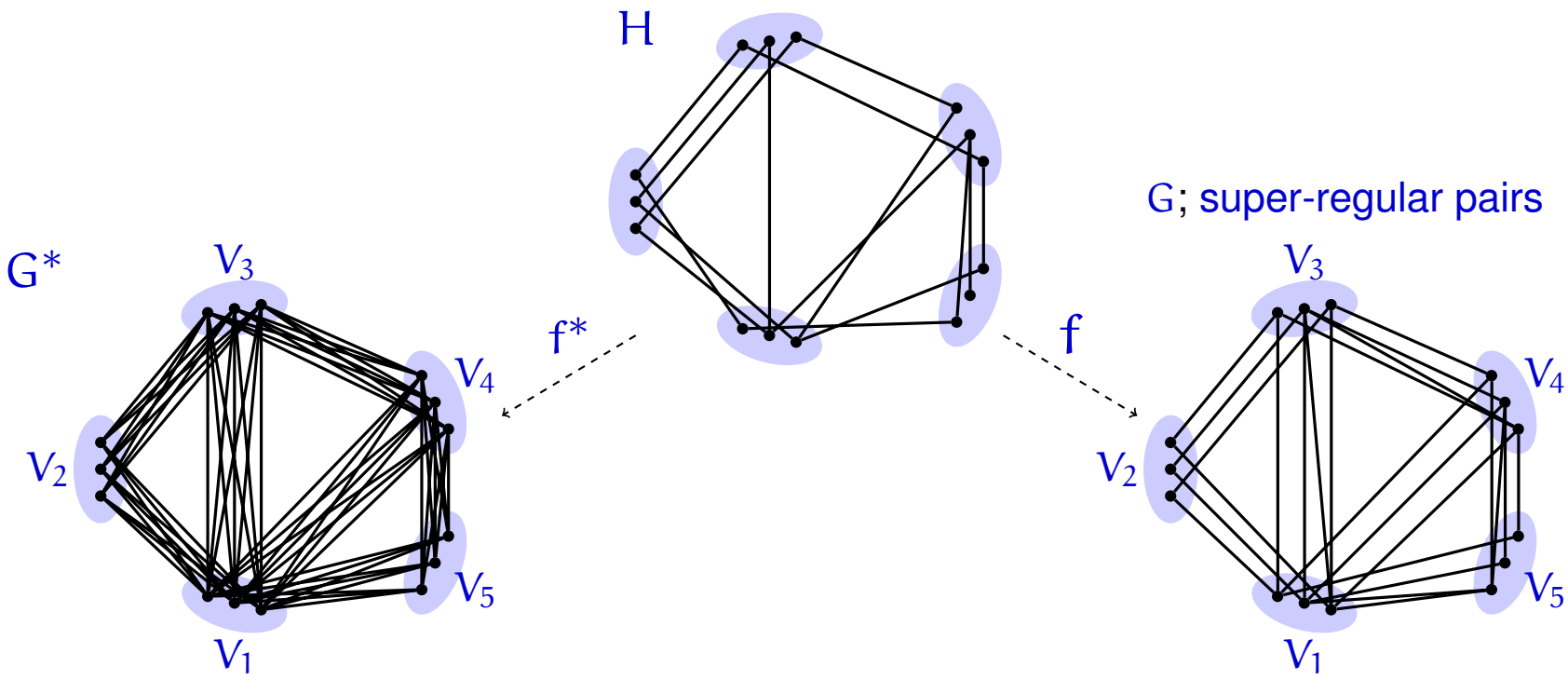
The Blow-up Lemma

Theorem 12 (The Blow-up Lemma (Komlós, Sárközy & Szemerédi 1997))**.** $\forall \delta > 0$, Δ, $r \in \mathbb{N}$ $\exists \varepsilon > 0$ *for which the following holds. Let* $G^* = (V_1, \dots, V_r; E^*)$ *and* $G = (V_1, \dots, V_r; E)$ *be two* r*-partite graphs. Suppose* $R \subset \binom{[r]}{2}$ *is such that if* $ij \in R$*, then* (V_i, V_j) *is a complete bipartite graph in* G^* *and* (V_i, V_j) *is an* (ε, δ)*-super-regular pair in* G*. If* H *with* $\Delta(H) \leq \Delta$ *can be If* H *with* $\Delta(H) \leq \Delta$ *can be* [. . .]

The Blow-up Lemma



The Blow-up Lemma



If f^* exists, the f exists.

The Blow-up Lemma

Theorem 13 (The Blow-up Lemma (Komlós, Sárközy & Szemerédi 1997)). $\forall \delta > 0$, Δ, $r \in \mathbb{N}$ $\exists \varepsilon > 0$ *for which the following holds. Let* $G^* = (V_1, \ldots, V_r; E^*)$ *and* $G = (V_1, \ldots, V_r; E)$ *be two* r*-partite graphs. Suppose* $R \subset \binom{[r]}{2}$ *is such that if* $ij \in R$*, then* (V_i, V_j) *is a complete bipartite graph in* G^* *and* (V_i, V_j) *is an* (ε, δ)*-super-regular pair in* G*. If* H *with* $\Delta(H) \leq \Delta$ *can be embedded into* G^**, then it can also be embedded into* G*.*

Remarks

- Already interesting in the case in which

$$G^* = (V_1^*, \dots, V_r^*; E^*) \quad \text{and} \quad |V_i^*| = .99|V_i| \text{ for all } i.$$

- Super-regularity not required in this case.
- Recall the challenge from Lecture I (almost spanning embedding lemma).

Proof of the blow-up lemma (rough sketch)

(*i*) Almost spanning embedding lemma

(*ii*) A König–Hall argument

Shall prove (*i*). Shall only mumble a few words about (*ii*).

Almost spanning embedding lemma

Blow-up lemma with $G^* = (V_1^*, \dots, V_r^*; E^*)$ and $|V_i^*| = .99|V_i|$ for every i. Suppose H has vertex classes $X_1, \dots, X_r$.

A trick (Alon–Füredi trick):

- ▷ Make the X_i 2-independent (or even more).
- ▷ Achieve this by refining the partition (X_i): use a proper colouring of the square of H.
- ▷ Can use Hajnal–Szemerédi (complementary form) to make the new vertex classes of H of the same size (up to 1).
- ▷ Refine the partition (V_i) randomly, watching out for the sizes (want the vertex partitions of H and G to be *compatible*).

Almost spanning embedding lemma

Suppose:

(*i*) X_i 2-independent in H and $|X_i| = (1-\mu)m$ for all $1 \le i \le r$.

(*ii*) $|V_i| = m$ for all i.

(*iii*) Set aside $V_i^q \subset V_i$ with $|V^q| = (\mu/2)m$ for each i.

(*iv*) $x_1, \ldots, x_\ell$ is some ordering, say τ, of $X = V(H)$.

(*v*) Shall embed the x_i basically following τ.

(*vi*) Partial embeddings will be $\psi_0 = \emptyset, \psi_1, \psi_2, \ldots$.

Almost spanning embedding lemma

▷ We'll map X_i into V_i. If $x \in X_i$, then let $V(x) = V_i$.

Further details: for $t = 0, 1, \ldots$, we let

(*i*) $C_t(x) = V(x) \cap \bigcap \{N_G(\psi_t(y)) : y \in N_H(x) \text{ and } y \in \mathrm{Dom}(\psi_t)\}$,

(*ii*) $A_t(x) = C_t(x) \setminus \mathrm{Im}(\psi_t)$,

(*iii*)

$$B_t(x) = \{v \in A_t(x) : \exists y \in N_H(x),\ y \notin \mathrm{Dom}(\psi_t),\ \text{and } |N_G(v) \cap A_t(y) \setminus V^q| < (\rho - \varepsilon)|A_t(y) \setminus V^q| \text{ or } |N_G(v) \cap C_t(y) \cap V^q| < (\rho - \varepsilon)|C_t(y) \cap V^q|\},$$

(*iv*) $\pi_t(x) = |\mathrm{Dom}(\psi_t) \cap N_H(x)|$.

Algorithm 1: Random greedy algorithm

Input : G and H with compatible partitions; an ordering τ on $X = V(H)$

$t := 0$; $\psi_0 := \emptyset$; $Q_0 := \emptyset$;
repeat
  let $x \in X \setminus (\mathrm{Dom}(\psi_t) \cup Q_t)$ be the next vertex in the order τ;
  choose $v \in A_t(x) \setminus B_t(x) \setminus V^q$ uniformly at random;
  $\psi_{t+1} := \psi_t \cup \{x \mapsto v\}$; $Q_{t+1} := Q_t$;
  foreach $y \in X \setminus \mathrm{Dom}(\psi_{t+1})$ **do**
    if $|A_{t+1}(y) \setminus V^q| < \frac{1}{2}\mu(p-\varepsilon)^{\pi_{t+1}(y)}|V(y)|$ **then**
      $Q_{t+1} := Q_{t+1} \cup \{y\}$;
  $t := t+1$;
until $\mathrm{Dom}(\psi_t) \cup Q_t = X$;
embed $H[Q_t]$ into $G[V^q]$, respecting ψ_t built above;

Embedding queue vertices

Let t_{RGAend} be the last value of t, so that we have the partial embedding $\psi_{t_{RGAend}}$ before we try to embed $H[Q_t]$ into $G[V^q]$. A crucial lemma is as follows:

Lemma 14. *With high probability, $Q_t \cap X_i$ never gets too large.*

If $|Q_t \cap X_i| \ll \rho^{\Delta}|V^q|$, then $H[Q_t] \subset G[V^q]$ (positive proportion embedding lemma).

Corollary 15. *The embedding of $H[Q_t]$ into $G[V^q]$ succeeds with high probability.*

The main lemma

Lemma 16 (Uniform distribution of candidate sets)**.** *Suppose* $\varepsilon \ll \mu, \rho, \kappa$ *and* $1/\Delta$ *and let* $\emptyset = \psi_0, \psi_1, \ldots, \psi_T$ *be partial embeddings with each* ψ_t *obtained from* ψ_{t-1} *by embedding some* $x \in V(H) \setminus \mathrm{Dom}(\psi_{t-1})$ *to a uniform random vertex from a subset of* $C_{t-1}(x)$ *of size at least*

$$\tfrac{1}{4}\mu(\rho - \varepsilon)^{\pi_{t-1}(x)}|V(x)|.$$

The following holds with probability at least $1 - r2^{-n/r}$*. For every* $i \in [r]$ *and every set* $W \subset V_i$ *of size at least* $\kappa|V_i|$*, the number of vertices* $x \in X_i$ *such that there exists* $t = t(x)$ *when* x *is unembedded and, moreover,*

$$|C_t(x) \cap W| < (\rho - \varepsilon)^{\pi_t(x)}\,|W|$$

is at most $\kappa|X_i|$*.*

The queues never get too large

Proof of Lemma 14 (rough sketch). Let $W = V_i \setminus V^q \setminus \mathrm{Im}(\psi_{t_{\mathrm{RGAend}}})$. Have $|W| \geq (\mu/2)|V_i|$. Suppose $x \in X_i$ got into Q_t at time $t = t(x)$. We have

$$C_t(x) \cap W \subset A_t(x) \setminus V^q.$$

Since $x \in Q_t$, we have

$$|C_t(x) \cap W| \leq |A_t(x) \setminus V^q| \leq \tfrac{1}{2}\mu(\rho - \varepsilon)^{\pi_t(x)}|V(x)| \leq (\rho - \varepsilon)^{\pi_t(x)}|W|.$$

By Lemma 16, have at most $\kappa|X_i|$ such x, with high probability. □

The main lemma

Proof of Lemma 16 (rough sketch). Fix i and $W \subset V_i$ with $|W| \geq \kappa|V_i|$. Suppose $x \in X_i$ is unembedded and, moreover,

$$|C_t(x) \cap W| < (\rho - \varepsilon)^{\pi_t(x)} |W|.$$

Pick the first time $t = t(x)$ such that the inequality above holds. Some $y \in N_H(x)$ was embedded to form ψ_t. This y was mapped onto a vertex v that is bad with respect to W (at most $\varepsilon|V(y)|$ choices), and v was selected from a set of size at least $\frac{1}{4}\mu(\rho - \varepsilon)^{\pi_{t-1}(y)}|V(y)|$. Probability of this event is $\lesssim \varepsilon/\mu\rho^{\Delta}$. Because of the 2-independence of X_i, the map $x \mapsto y(x)$ is injective. The probabilities multiply, and the probability that there are $\kappa|X_i|$ such x is

$$\lesssim \left(\frac{\varepsilon\Delta}{\mu\rho^{\Delta}}\right)^{\kappa|X_i|} \leq 2^{-4|X_i|}.$$

Now use union bound over all i and W. □

The König–Hall argument

(*i*) Now $G^* = (V_1^*, \dots, V_r^*; E^*)$ and $|V_i^*| = |V_i|$ for all i.

(*ii*) **Key:** set aside a *set of buffer vertices* in each X_i, say, $X_i^{\mathrm{buff}} \subset X_i$. Let $|X_i^{\mathrm{buff}}| = \mu|X_i|$.

(*iii*) Choose X_i^{buff} so that $X^{\mathrm{buff}} = \bigcup_i X_i^{\mathrm{buff}}$ is 4-independent.

(*iv*) Run the random greedy algorithm between $X \setminus X^{\mathrm{buff}}$ and $V = V(G)$. This gives $\psi^- \colon H[X \setminus X^{\mathrm{buff}}] \hookrightarrow G$.

(*v*) Embed X^{buff} into $V^{\mathrm{buff}} = V \setminus \mathrm{Im}(\psi^-)$ by showing that $C(x) \cap V^{\mathrm{buff}}$ ($x \in X^{\mathrm{buff}}$) has a system of distinct representatives $(v_x)_{x \in X^{\mathrm{buff}}}$. Extend ψ^- by mapping x onto v_x for all $x \in X^{\mathrm{buff}}$.

(*vi*) Check König–Hall conditions: small sets (easy), medium sized sets (harder), very large sets (hard).

Example applications

(*i*) Very approximate Alon–Yuster

(*ii*) Square of almost Hamilton cycles—approximate version

Very approximate Alon–Yuster

Theorem 17. *Fix* $\varepsilon > 0$ *and* $H = H^h$*. Then any* $G = G^n$ *with* $n \geq n_0(\varepsilon, H)$ *with minimum degree* $\delta(G) \geq (1 - 1/\chi(H) + \varepsilon)n$ *contains* $(1 - \varepsilon)n/h$ *vertex disjoint copies of* H*.*

- Doesn't require blow-up; constant size embedding lemma suffices.

Useful to note that the clean-up graph G^* obtained from regularization may be required to satisfy a minimum degree condition.

Cleaned-up graph G^* with minimum degree condition

Definition 18. *After regularization of* G*, have* $V = V_1 \cup \cdots \cup V_t$*. Remove all edges in* $G[V_i, V_j]$ *for all* i *and* j *such that*

1. (V_i, V_j) *is not* ε*-regular,*
2. $|E(V_i, V_j)| \leq f(\varepsilon)|V_i||V_j|$ (*suitable* f *with* $f(\varepsilon) \to 0$ *as* $\varepsilon \to 0$)*.*

Resulting graph: cleaned-up graph G^**.*

In G^*, every $G^*[V_i, V_j]$ is regular and 'dense'. Usually, lose very little. By allowing an exceptional class V_0 with $|V_0| \leq \varepsilon|V|$, may require, for every $v \in V(G) \setminus V_0$, that

$$\deg_{G^*}(v) \geq \deg_G(v) - g(\varepsilon)n,$$

for some g with $g(\varepsilon) \to 0$ as $\varepsilon \to 0$.

Very approximate Alon–Yuster

Exercise 19. *Prove the very approximate Alon–Yuster theorem.*

Hint 20. *Clean-up graph with mindeg condition + Hajnal–Szemerédi.*

Squares of almost Hamilton cycles—approximate form

Theorem 21. *Suppose* $G = G^n$ *and* $n \geq n_0$. *If* $\delta(G) \geq (2/3 + \varepsilon)n$, *then* G *contains the square of a cycle with* $\geq (1 - \varepsilon)n$ *vertices.*

Exercise 22. *Prove the theorem above.*

Hint 23. *Clean-up graph with mindeg condition* $+$ *Corrádi–Hajnal* $+$ *almost spanning blow-up lemma* $+$ *image restrictions.*

Summary

- Got a glimpse into how to use the regularity method to find large subgraphs in given graphs; key tool: **blow-up lemma**.

The regularity method and blow-up lemmas for sparse graphs

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Course outline

1. **Lecture I.** Introduction to the regularity method
2. **Lecture II.** The blow-up lemma
3. **Lecture III.** The sparse case: small subgraphs
4. **Lecture IV.** The sparse case: large subgraphs

Lecture III. The sparse case (small subgraphs)

Motivation

- Shall have a fairly long discussion on motivation.
- Shall discuss rudiments of the regularity method for sparse graphs.

Restricted Ramsey theory

- ▷ We know $K^6 \to K^3$.
- ▷ Are there G with $K^6 \not\subset G$ such that $G \to K^3$?
- ▷ Yes: $G = K^8 \setminus C^5 = C^3 \vee C^5$ (Graham 1968)
- ▷ Also $G = (K^9 \setminus C^9) \vee K^1$ (Leader)

Definition 1. *For positive integers* r, k *and* ℓ, *let*

$$f(r,k,\ell) = \min\left\{n\colon \exists G = G^n \text{ such that } G \not\supset K^\ell \text{ and } G \to (K^k)_r\right\}.$$

- ◦ $G \to (H)_r$: Ramsey property for r-colouring of the edges

Restricted Ramsey theory

Definition 2. *For positive integers* r, k *and* ℓ, *let*

$$f(r,k,\ell) = \min\left\{n\colon \exists G = G^n \text{ such that } G \not\supset K^\ell \text{ and } G \to (K^k)_r\right\}.$$

- $G \to (H)_r$: Ramsey property for r-colouring of the edges

Question 3 (Erdős and Hajnal 1967)**.** *Is* $f(r,k,\ell) < \infty$ *for every positive* r, k *and* ℓ *with* $k < \ell$*?*

Answer. Yes: Folkman 1970, Nešetřil and Rödl 1976

Numbers. Graham: $f(2,3,6) = 8$. Nenov, Piwakowski, Radziszowski and Urbański: $f(2,3,5) = 15$. How about $f(2,3,4)$?

Erdős: $f(2,3,4) \leq 10^{10}$?

Restricted Ramsey theory

Erdős: $f(2,3,4) \leq 10^{10}$?

(*i*) Lu 2008: $f(2,3,4) \leq 9697$

(*ii*) Dudek and Rödl 2008: $f(2,3,4) \leq 941$

Let us go back to the proof of $f(r,k,\ell) < \infty$.

Natural approach today. Investigate whether can use random graphs.

The random graphs $G(n,p)$

- $G(n,p)$: each element of $\binom{[n]}{2}$ is present with probability p, independently of all others

Always interested in $n \to \infty$. Use the terms 'almost surely', 'almost every', 'almost always', etc to mean 'with probability $\to 1$ as $n \to \infty$'.

- We are interested in the existence of monoχ subgraphs in **coloured random graphs**; i.e., properties of the form $G(n,p) \to (H)_r$.
- Also interested in existence of monochromatic subgraphs in 'dense' subgraphs of random graphs; i.e., statements of the form $G(n,p) \to_\eta H$ (any η-fraction of the edges of $G(n,p)$ contains a copy of H).

The results for K^3

Theorem 4. *There is a large enough constant* C *such that if* $p \geq C/\sqrt{n}$, *then a.e.* $G(n,p)$ *satisfies*

$$G(n,p) \to (K^3)_2.$$

Theorem 5. *For any* $\eta > 0$, *there* C *such that if* $p \geq C/\sqrt{n}$ *then a.e.* $G(n,p)$ *satisfies*

$$G(n,p) \to_{1/2+\eta} K^3.$$

$f(2,3,4) < \infty$

(*i*) Basic remark: $G(n, C/\sqrt{n})$ a.s. contains $O(p^6 n^4) = O(n)$ copies of K^4.

(*ii*) Two approaches:

- ▷ Show that if we delete any $O(n)$ edges from $G(n,p)$ still have a Ramsey graph for K^3.
- ▷ Show that the number of monoχ K^3 is large (it is actually $(1/4 + o(1))p^3\binom{n}{3}$ a.s.) and that an edge does not belong to too many K^3s.

(*iii*) It follows that $f(2,3,4) < \infty$.

The case $p = 1$ (skip/exercise)

Theorem 6 (Goodman)**.** *The number of monochromatic triangles in any 2-colouring of* K^n *is* $\geq \binom{n}{3} - \frac{1}{8}n(n-1)^2 = (\frac{1}{4} + o(1))\binom{n}{3}$.

Proof. Count the number N of 2-coloured cherries. Say have $b(x)$ blue edges and $r(x)$ red edges at vertex x. Have $N = \sum_x b(x)r(x) \leq n(n-1)^2/4$ (use $b(x) + r(x) = n - 1$). The number of non-monochromatic triangles is $N/2$. □

Clearly, there are 2-colourings with $\leq \frac{1}{4}\binom{n}{3}$ monochromatic triangles!

The case $p = 1$ (skip/exercise)

Theorem 7. *Goodman implies Mantel:* $\mathrm{ex}(n, K^3) \le n^2/4$.

Proof. Suppose $K^3 \not\subset G^n$ and $e(G^n) > n^2/4$, so that $e(G^n) \ge n^2/4 + 3/4$. Count the number N of pairs (e, T) where $e \in E(G^n)$, $T \in \binom{V(G^n)}{3}$ and $e \subset T$. Say there are t_i triples containing i edges of G. Then, by Theorem 6,

$$\begin{aligned}(n^2/4 + 3/4)(n-2) &\le e(G^n)(n-2) \\ &= N \le t_1 + 2t_2 \le 2(t_1 + t_2) \le n(n-1)^2/4,\end{aligned}$$

which is a contradiction for $n \ge 4$. □

Theorems 4 and 5 from Goodman's counting (skip/exercise)

Theorem 8. *For every $\varepsilon > 0$ there is C such that if $p \geq C/\sqrt{n}$, then a.e. $G(n,p)$ is such that any 2-colouring of its edges contains at least $(1/4 - \varepsilon)p^3\binom{n}{3}$ monochromatic K^3.*

Proof. Exercise! □

Exercise 9. *Derive Theorem 5 from Theorem 8.*

Exercise 10. *Show that the threshold for Theorem 4 is indeed $1/\sqrt{n}$.*

Exercise 11. *Show that the threshold for Theorem 5 is indeed $1/\sqrt{n}$.*

How about $f(r, k, \ell)$?

We are interested in the *threshold* for the Ramsey property

$$G(n, p) \to (K^k)_r. \tag{1}$$

Threshold: the 'smallest' order of growth for $p = p(n)$ such that (1) is guaranteed to hold a.s.

This was actually a very natural problem in the theory of random graphs, and was solved by Rödl and Ruciński (1995).

The Rödl–Ruciński theorem

Definition 12 (2-density and $m_2(H)$)**.** *The 2-density* $d_2(H)$ *of a graph* H *with* $|V(H)| > 2$ *is*

$$\frac{|E(H)| - 1}{|V(H)| - 2}. \tag{2}$$

For $H = K^1$ *and* $2K^1$ *let* $d_2(H) = 0$*; set* $d_2(K^2) = 1/2$*. Let*

$$m_2(H) = \max\{d_2(J) \colon J \subset H,\ |V(J)| > 0\}. \tag{3}$$

Exercise 13. *Consider, say,* $H = K^h$*. Show that if* $p \ll n^{-1/m_2(H)}$*, then a.s.* $\#\{H \hookrightarrow G(n,p)\} \ll e(G(n,p))$*. On the other hand, if* $p \gg n^{-1/m_2(H)}$*, then a.s.* $\#\{H \hookrightarrow G(n,p)\} \gg e(G(n,p))$*.*

The Rödl–Ruciński theorem

Theorem 14. *Let H be a graph containing at least a cycle and let $r \geq 2$ be an integer. Then there exist constants c and C such that*

$$\lim_{n\to\infty} \mathbb{P}(G(n,p) \to (H)_r) = \begin{cases} 0 & \text{if } p \leq cn^{-1/m_2(H)} \\ 1 & \text{if } p \geq Cn^{-1/m_2(H)}. \end{cases} \tag{4}$$

- In particular, $p_0 = p_0(n) = n^{-1/m_2(H)}$ is the threshold for the property $G(n,p) \to (H)_r$.

The Rödl–Ruciński theorem

Exercise 15. *Prove that* $f(r, k, \ell) < \infty$ *whenever* $k < \ell$.

Exercise 16. *Given a graph* G, *let* $\mathcal{H}_3(G)$ *be the* 3*-uniform hypergraph whose hypervertices are the edges of* G *and the hyperedges are the edge sets of the triangles in* G. *Show that, for any integers* ℓ *and* r, *there is a graph* G *satisfying* $G \to (K^3)_r$ *such that* $\mathcal{H}_3(G)$ *has girth* $\geq \ell$.

Turán type results for subgraphs of random graphs

Generalized Turán number:

$$\mathrm{ex}(G, H) = \max\left\{|E(G')|\colon H \not\subset G' \subset G\right\}. \tag{5}$$

- $\mathrm{ex}(n, H) = \mathrm{ex}(K^n, H)$
- $\mathrm{ex}(Q^d, C^4) = ?$
- $\mathrm{ex}(G, K^h) = ?$ for (n, d, λ)-graphs G

Turán type results for subgraphs of random graphs

Exercise 17. *Show that, for any* G *and* H*, we have*

$$\mathrm{ex}(G,H) \geq \left(1 - \frac{1}{\chi(H)-1}\right) e(G). \tag{6}$$

▷ Interested in knowing when this is sharp for $G = G(n,p)$ (up to $o(e(G))$)

Turán type results for subgraphs of random graphs

Theorem 18 (K., Rödl and Schacht 2004)**.** *Let* H *be a graph with degeneracy* d *and suppose* $np^d \gg 1$*. Then, almost surely,*

$$\mathrm{ex}(G(n,p),H) = \left(1 - \frac{1}{\chi(H)-1} + o(1)\right) p\binom{n}{2}. \tag{7}$$

Conjecture 19. *For any graph* H*, if* $np^{m_2(H)} \to \infty$ *then* (7) *holds almost surely.*

Example: if $H = K^k$, have $m_2(H) = (k+1)/2$, but Theorem 18 supposes $np^{1/(k-1)} \gg 1$.

Turán type results for subgraphs of random graphs

Conjecture 19 was proved in 2009/2010:

Theorem 20 (Schacht 2016; Conlon and Gowers 2016)**.** *Conjecture 19 holds.*

- Quite different approaches:
  - Schacht: combinatorial, multi-round exposure method
  - Conlon and Gowers: analytic approach
  - **General transference principles**

General transference principles

- **Schacht** and **Conlon & Gowers** obtained general transference results: results that allow one to *transfer* classical density results ("full environment") to random sparse subenvironments
- **Applications:** results in additive combinatorics; graph and hypergraph theorems

The regularity method for sparse random graphs

- Early approaches to Conjecture 19: regularity method of Szemerédi for subgraphs of sparse random graphs
  - Schacht and Conlon and Gowers bypassed the regularity approach.
- Recently, the regularity method has been fully developed for subgraphs of random graphs: Balogh, Morris, Samotij (2015), Saxton and Thomason (2015), Conlon, Gowers, Samotij, and Schacht (2014).
- **In short:** *the regularity method of Szemerédi works in sparse random environments.*

Recall

- Shall have a fairly long discussion on motivation.
- Shall discuss rudiments of the regularity method for sparse graphs.

Szemerédi's regularity lemma

1. Tool for identifying the quasirandom structure of deterministic graphs.
2. Works very well for large, dense graphs: n-vertex graphs with $\geq cn^2$ edges, $n \to \infty$.
3. Variant for sparse graphs exists (sparse = with $o(n^2)$ edges).
4. With the recent embedding lemmas, the method works nicely/easily when dealing with small subgraphs.

ε-regularity

Definition 21 (ε-regular pair)**.** $G = (V, E)$ *a graph;* $U, W \subset V$ *non-empty and disjoint. Say* (U, W) *is* ε*-regular* (*in* G) *if*

- *for all* $U' \subset U$, $W' \subset W$ *with* $|U'| \geq \varepsilon|U|$ *and* $|W'| \geq \varepsilon|W|$, *we have*

$$\left| \frac{|E(U', W')|}{|U'||W'|} - \frac{|E(U, W)|}{|U||W|} \right| \leq \varepsilon. \tag{8}$$

ε-regularity

The pair (U, W) is ε-regular if

$$\left| \frac{|E(U', W')|}{|U'||W'|} - \frac{|E(U, W)|}{|U||W|} \right| \leq \varepsilon. \tag{9}$$

Equivalently,

$$|E(U', W')| = |U'||W'| \left(\frac{|E(U, W)|}{|U||W|} \pm \varepsilon \right). \tag{10}$$

Clearly, not meaningful if

$$\frac{|E(U, W)|}{|U||W|} \to 0 \tag{11}$$

and ε is fixed. (We think of $G = (V, E)$ with $n = |V| \to \infty$.)

ε-regularity; multiplicative error version

Replace

$$|E(U', W')| = |U'||W'| \left(\frac{|E(U, W)|}{|U||W|} \pm \varepsilon \right) \tag{12}$$

by

$$|E(U', W')| = (1 \pm \varepsilon)|E(U, W)| \frac{|U'||W'|}{|U||W|}. \tag{13}$$

Altered condition becomes

▷ *for all* $U' \subset U$, $W' \subset W$ *with* $|U'| \geq \varepsilon|U|$ *and* $|W'| \geq \varepsilon|W|$, *we have*

$$\left| |E(U', W')| - |E(U, W)| \frac{|U'||W'|}{|U||W|} \right| \leq \varepsilon |E(U, W)| \frac{|U'||W'|}{|U||W|}. \tag{14}$$

OK even if $|E(U, W)|/|U||W| \to 0$.

Szemerédi's regularity lemma, sparse version

Any graph with no 'dense patches' admits a Szemerédi partition with the new notion of ε-regularity.

Definition 22 ((η, b)-bounded)**.** *Say* $G = (V, E)$ *is* (η, b)*-bounded if for all* $U \subset V$ *with* $|U| \geq \eta|V|$*, we have*

$$\#\{\textit{edges within } U\} \leq b|E|\binom{|U|}{2}\binom{|V|}{2}^{-1}. \tag{15}$$

Szemerédi's regularity lemma, sparse version

Theorem 23 (The regularity lemma)**.** *For any $\varepsilon > 0$, $t_0 \geq 1$, and b, there exist $\eta > 0$ and T_0 such that any (η, b)-bounded graph G admits a partition $V = V_1 \cup \cdots \cup V_t$ such that*

1. $|V_1| \leq \cdots \leq |V_t| \leq |V_1| + 1$
2. $t_0 \leq t \leq T_0$
3. *at least $(1-\varepsilon)\binom{t}{2}$ pairs (V_i, V_j) $(i < j)$ are ε-regular.*

Proof. Just follow Szemerédi's original proof. □

A counting lemma (simplest version)

Setup. $G = (V_1, V_2, V_3; E)$ tripartite with

1. $|V_i| = m$ for all i
2. (V_i, V_j) ε-regular for all $i < j$
3. $|E(V_i, V_j)| = \rho m^2$ for all $i < j$

Notation: $G = G_3^{(\varepsilon)}(m, \rho)$ [G is an ε-regular triple with density ρ]

$\triangleright$ Wish to embed K^3 with $V(K^3) = \{x_1, x_2, x_3\}$ such that x_i is placed in V_i.

A counting lemma (simplest version)

Just like random:

Lemma 24 (Counting Lemma; Embedding lemma)**.** $\forall \rho > 0$, $\delta > 0$ $\exists \varepsilon > 0$, m_0: *if* $m \geq m_0$, *then*

$$\left| \#\{K^3 \hookrightarrow G_3^{(\varepsilon)}(m, \rho)\} - \rho^3 m^3 \right| \leq \delta m^3. \tag{16}$$

Tough: counting Lemma is false if $\rho \to 0$

Fact 25. $\forall \varepsilon > 0\ \exists \rho > 0,\ m_0\ \forall m \geq m_0\ \exists G_3^{(\varepsilon)}(m, \rho)$ *with*

$$K^3 \not\subset G_3^{(\varepsilon)}(m, \rho). \tag{17}$$

[cf. Lemma 24]

Exercise 26. *Prove Fact 25.*

Hint 27. *Consider the random tripartite graph* $G(t, t, t; \delta/\sqrt{t})$ ($t = t(\varepsilon)$ *and* $\delta = \delta(\varepsilon)$ *appropriate constants*). *Remove few edges to make it* K^3*-free. Blow up each vertex to an independent set of size* n/t.

Change of focus (just for simplicity): from *counting* $K^3 \subset G_3^{(\varepsilon)}(m, \rho)$ to *existence of* $K^3 \subset G_3^{(\varepsilon)}(m, \rho)$.

Key observation

Counterexamples to the embedding lemma in the sparse setting do exist (Fact 25), but

are extremely rare.

An asymptotic enumeration lemma

Lemma 28. $\forall \beta > 0\ \exists \varepsilon > 0, C > 0, m_0$: *if* $T = \rho m^2 \geq C m^{3/2}$, *then*

$$\#\{G_3^{(\varepsilon)}(m,\rho) \not\supset K^3\} \leq \beta^T \binom{m^2}{T}^3. \tag{18}$$

Observe that $\rho \geq C/\sqrt{m} \to 0$.

Consequence for random graphs

Easy expectation calculations imply

- ▷ if $p \gg 1/\sqrt{n}$, then almost every $G(n,p)$ is such that

$$\left(K^3\text{-free } G_3^{(\varepsilon)}(m,\rho)\right) \not\subset G(n,p), \tag{19}$$

  if (*) $mp \gg \log n$ and $\rho \geq \alpha p$ for some fixed α.

Conclusion. Recovered an 'embedding lemma' in the sparse setting, *for subgraphs of random graphs*.

Corollary 29 (EL for subgraphs of r.gs)**.** *If* $p \gg 1/\sqrt{n}$ *and* (*) *holds, then almost every* $G(n,p)$ *is such that if* $G_3^{(\varepsilon)}(m,\rho) \subset G(n,p)$*, then*

$$\exists \iota\colon K^3 \hookrightarrow G_3^{(\varepsilon)}(m,\rho) \subset G(n,p). \tag{20}$$

The easy calculation

Recall $T = \rho m^2$, $\rho \geq \alpha p$, and $mp \gg \log n$. Therefore

$$\mathbb{E}(\#\{K^3\text{-free } G_3^{(\varepsilon)}(m,\rho) \hookrightarrow G(n,p)\}) \leq n^{3m} o(1)^T \binom{m^2}{T}^3 p^{3T}$$

$$\leq n^{3m} \left(o(1)\frac{em^2}{T}\right)^{3T} p^{3T} \leq n^{3m} \left(o(1)\frac{em^2 p}{T}\right)^{3T}$$

$$= n^{3m} \left(o(1)\frac{em^2 p}{\rho m^2}\right)^{3T} \leq n^{3m} \left(o(1)\frac{e}{\alpha}\right)^{3T}$$

$$\leq e^{3m\log n} \left(o(1)\frac{e}{\alpha}\right)^{3\alpha m \log n} = o(1).$$

Superexponential bounds

Suppose we wish to prove a statement about all subgraphs of $G(n,p)$.

- Too many such subgraphs: about $2^{p\binom{n}{2}}$
- $G(n,p)$ has no edges with probability $(1-p)^{\binom{n}{2}} \geq \exp\{-2pn^2\}$, if, say, $p \leq 1/2$.
- Concentration inequalities won't do ($2^{p\binom{n}{2}}$ vs e^{-2pn^2}).
- Bounds of the form

$$o(1)^T\binom{\binom{m}{2}}{T} \tag{21}$$

for the cardinality of a family of 'undesirable subgraphs' $U(m,T)$ do the job. Use of such bounds goes back to Füredi (1994).

An embedding lemma for K^3 (sparse setting)

Scheme:

1. Proved an asymptotic enumeration lemma:

$$\#\{G_3^{(\varepsilon)}(m,\rho) \not\supset K^3\} = o(1)^T \binom{m^2}{T}^3. \tag{22}$$

   [$T = \rho m^2$]

2. Observed that this implies a.e. $G(n, \omega/\sqrt{n})$ contains no K^3-free $G_3^{(\varepsilon)}(m,\rho)$ [any $\omega = \omega(n) \to \infty$ as $n \to \infty$].

3. Obtained a K^3-embedding lemma for subgraphs of $G(n,p)$, even when $p = \omega/\sqrt{n}$: for all $G_3^{(\varepsilon)}(m,\rho) \subset G(n,p)$, have $K^3 \subset G_3^{(\varepsilon)}(m,\rho)$.

General graphs H?

Suppose $|V(H)| > 2$. Recall

$$d_2(H) = \frac{|E(H)| - 1}{|V(H)| - 2} \tag{23}$$

and

$$m_2(H) = \max\{d_2(J) \colon J \subset H\}. \tag{24}$$

Theorem 30 (Balogh, Morris & Samotij 2015 and Saxton & Thomason 2015)**.** $\forall H$, $\beta > 0$ $\exists \varepsilon > 0$, $C > 0$, m_0: *if* $T = \rho m^2 \geq C m^{2-1/m_2(H)}$, *then*

$$\#\{G_H^{(\varepsilon)}(m,\rho) \not\supset H\} \leq \beta^T \binom{m^2}{T}^{e(H)}. \tag{25}$$

▷ Can't miss: container lectures by Rob Morris next week!

Consequences of Theorem 30

1. The Rödl–Ruciński theorem on the threshold for Ramsey properties of random graphs (1-statement) and the Turán counterpart, with the best possible edge probability.

2. Łuczak (2000): structural and enumerative consequences for H-free graphs on n vertices and M edges

Consequences of Theorem 30

Theorem 31 (Łuczak 2000)**.** *Suppose* $\chi(H) = h \geq 3$*. Then for every* $\delta > 0$ *there exists* $C = C(\delta, H)$ *such that, almost surely, a graph chosen uniformly at random from the family of all* H*-free labelled graphs on* n *vertices and* $M \geq Cn^{2-1/m_2(H)}$ *edges can be made* $(h-1)$*-partite by removing* $\leq \delta M$ *edges.*

Consequences of Theorem 30

Theorem 32 (Łuczak 2000)**.** *Suppose* $\chi(H) = h \geq 3$*. Then for every* $\varepsilon > 0$ *there exist* $C = C(\varepsilon, H)$ *and* $n_0 = n_0(\varepsilon, H)$ *such that, for* $n \geq n_0$ *and* $Cn^{2-1/d_2(H)} \leq M \leq n^2/C$*, we have*

$$\left(\frac{h-2}{h-1} - \varepsilon\right)^M \leq \mathbb{P}\left(G(n, M) \not\supset H\right) \leq \left(\frac{h-2}{h-1} + \varepsilon\right)^M. \tag{26}$$

▷ $G(n, M)$: random graph of order n with M edges (uniform distribution)

The hereditary nature of regularity

Setup. $B = (U, W; E)$ an ε-regular bipartite graph with $|U| = |W| = m$ and $|E| = \rho m^2$, $\rho > 0$ constant, and an integer d. Sample $N \subset U$ and $N' \subset W$ with $|N| = |N'| = d$ uniformly at random.

Theorem 33 (Duke and Rödl 1985)**.** *For any $\beta > 0$, $\rho > 0$, and $\varepsilon' > 0$, if $\varepsilon \le \varepsilon_0(\beta, \rho, \varepsilon')$, $d \ge d_0(\beta, \rho, \varepsilon')$, and $m \ge m_0(\beta, \rho, \varepsilon')$, then*

$$\mathbb{P}\Big((N, N') \textit{ bad}\Big) \le \beta^d, \tag{27}$$

where (N, N') is bad if $\Big||E(N, N')|d^{-2} - \rho\Big| > \varepsilon'$ or else (N, N') is not ε'-regular.

The hereditary nature of regularity

Exercise 34. *For any k and $\delta > 0$, there is C such that the following holds. If $\chi(G - F) \geq k$ for any $F \subset E(G)$ with $|F| \leq \delta n^2$, then there is $H \subset G$ with $\chi(H) \geq k$ and $|V(H)| \leq C$. Can you guarantee many such 'witnesses' H?*

Local characterization for regularity

Setup. $B = (U, W; E)$, a bipartite graph with $|U| = |W| = m$. Consider the properties

(PC) for some constant p, have $m^{-1} \sum_{u \in U} |\deg(u) - pm| = o(m)$ and

$$\frac{1}{m^2} \sum_{u,u' \in U} |\deg(u,u') - p^2 m| = o(m). \tag{28}$$

(R) (U, W) is $o(1)$-regular (classical sense).

Theorem 35. (PC) *and* (R) *are equivalent.*

A proof of Theorem 33

Let a graph $F = (U, E)$ with $|U| = m$ and $|E| \leq \eta\binom{m}{2}$ be given. Suppose we select a d-set N uniformly at random from U. We are then interested in giving an upper bound for $e(F[N])$, the number of edges that the set N will induce in F.

Lemma 36. *For every α and $\beta > 0$, there exist $\eta_0 = \eta_0(\alpha, \beta) > 0$ such that, whenever $0 < \eta \leq \eta_0$, we have*

$$\mathbb{P}\left(e(F[N]) \geq \alpha\binom{d}{2}\right) \leq \beta^d. \tag{29}$$

Proof. Exercise! □

Exercise 37. *Use Lemma 36 and Theorem 35 to prove Theorem 33.*

The hereditary nature of sparse regularity

Definition 38 ((ε, p)-lower-regularity)**.** *Suppose* $0 < \varepsilon < 1$ *and* $0 < p \leq 1$*. A bipartite graph* $B = (U, W; E)$ *is* (ε, p)*-lower-regular if for all* $U' \subset U$ *and* $W \subset W$ *with* $|U'| \geq \varepsilon|U|$ *and* $|W'| \geq \varepsilon|W|$*, we have*

$$\frac{e(U', W')}{|U'||W|} \geq (1 - \varepsilon)p. \tag{30}$$

The hereditary nature of sparse regularity

Setup. $B = (U, W; E)$ an (ε, p)-lower-regular bipartite graph with $|U| = |W| = m$ and $|E| = pm^2$ and integer d. Sample $N \subset U$ and $N' \subset W$ with $|N| = |N'| = d$ uniformly at random.

Theorem 39 (Gerke, K., Rödl & Steger 2007)**.** *For all* $0 < \beta, \varepsilon' < 1$*, there exist* $\varepsilon_0 = \varepsilon_0(\beta, \varepsilon') > 0$ *and* $C = C(\varepsilon')$ *such that, for any* $0 < \varepsilon \leq \varepsilon_0$ *and* $0 < p < 1$*, the following holds. Let* $G = (U, W; E)$ *be an* (ε, p)*-lower-regular bipartite graph and suppose* $d \geq Cp^{-1}$*. Then*

$$\mathbb{P}\Big((N, N') \textit{ bad}\Big) \leq \beta^d, \tag{31}$$

where (N, N') *is bad if* (N, N') *is not* (ε', p)*-lower-regular.*

The hereditary nature of sparse regularity

Setup. $B = (U, W; E)$ an (ε, p)-lower-regular bipartite graph with $|U| = |W| = m$, $|E| = pm^2$, and $p \geq \alpha q$. Also, suppose we have two other bipartite graphs $A = (U', U; E_A)$ and $C = (W, W'; E_C)$, also (ε, p)-lower-regular.

Corollary 40 (Quite imprecise...). *Suppose* $A \cup B \cup C \subset G(n, q)$ *and* $u' \in U'$ *and* $w' \in W'$ *are 'typical' vertices. If* $pm \gg 1/p$ *then* $B[\Gamma_A(u'), \Gamma_C(w')]$ *is* $(f(\varepsilon), p)$*-lower-regular* $(f(\varepsilon) \to 0$ *as* $\varepsilon \to 0)$.

▷ Corollary above may be used in inductive embedding schemes.

Local characterization for sparse regularity

Remark: (very imprecisely) a similar statement to Theorem 35 may be proved for subgraphs of random graphs.

The regularity method and blow-up lemmas for sparse graphs

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Course outline

1. **Lecture I.** Introduction to the regularity method
2. **Lecture II.** The blow-up lemma
3. **Lecture III.** The sparse case: small subgraphs
4. **Lecture IV.** The sparse case: large subgraphs

Lecture IV. The sparse case (large subgraphs)

Aim of talk

- Shall only give some idea (shan't be ambitious)
- **Joint work with**
  - P. Allen (London),
  - J. Böttcher (London),
  - H. Hàn (Santiago),
  - Y. Person (Frankfurt).
- Preprint available on the school's webpage

Outline of talk

1. (Lecture I & II) The regularity method; the blow-up lemma
2. (Lecture III) The regularity method in the sparse setting
3. Applications of the blow-up lemmas in the sparse setting
4. Inheritance of regularity
5. Statement of the sparse blow-up lemmas

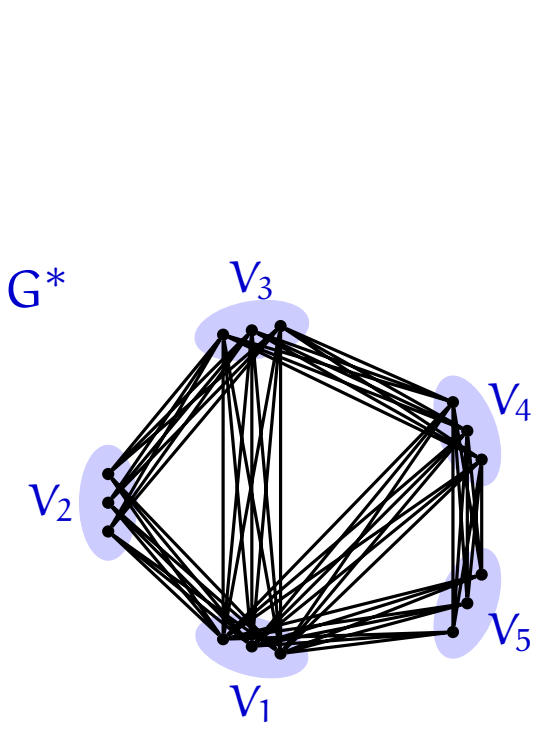
Outline of talk

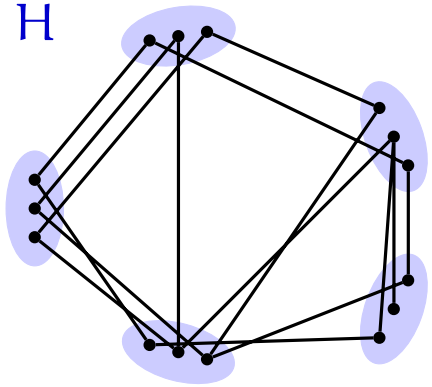
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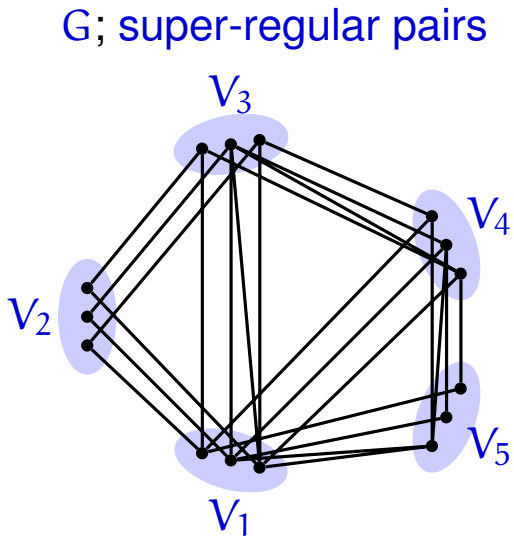
The Blow-up Lemma

Theorem 1 (The Blow-up Lemma (Komlós, Sárközy & Szemerédi '97))**.** *For every* $\delta > 0$, Δ, $r \in \mathbb{N}$ *there is* $\varepsilon > 0$ *such that the following holds. Let* $G^* = (V_1, \dots, V_r; E^*)$ *and* $G = (V_1, \dots, V_r; E)$ *be two graphs and let* $R \subset \binom{[r]}{2}$ *be such that* (V_i, V_j) *is a complete bipartite graph in* G^* *and* (V_i, V_j) *is an* (ε, δ)*-super-regular graph in* G *whenever* $ij \in R$*. If* H *with* $\Delta(H) \leq \Delta$ *can be embedded into* G^* *then it can also be embedded into* G*.*

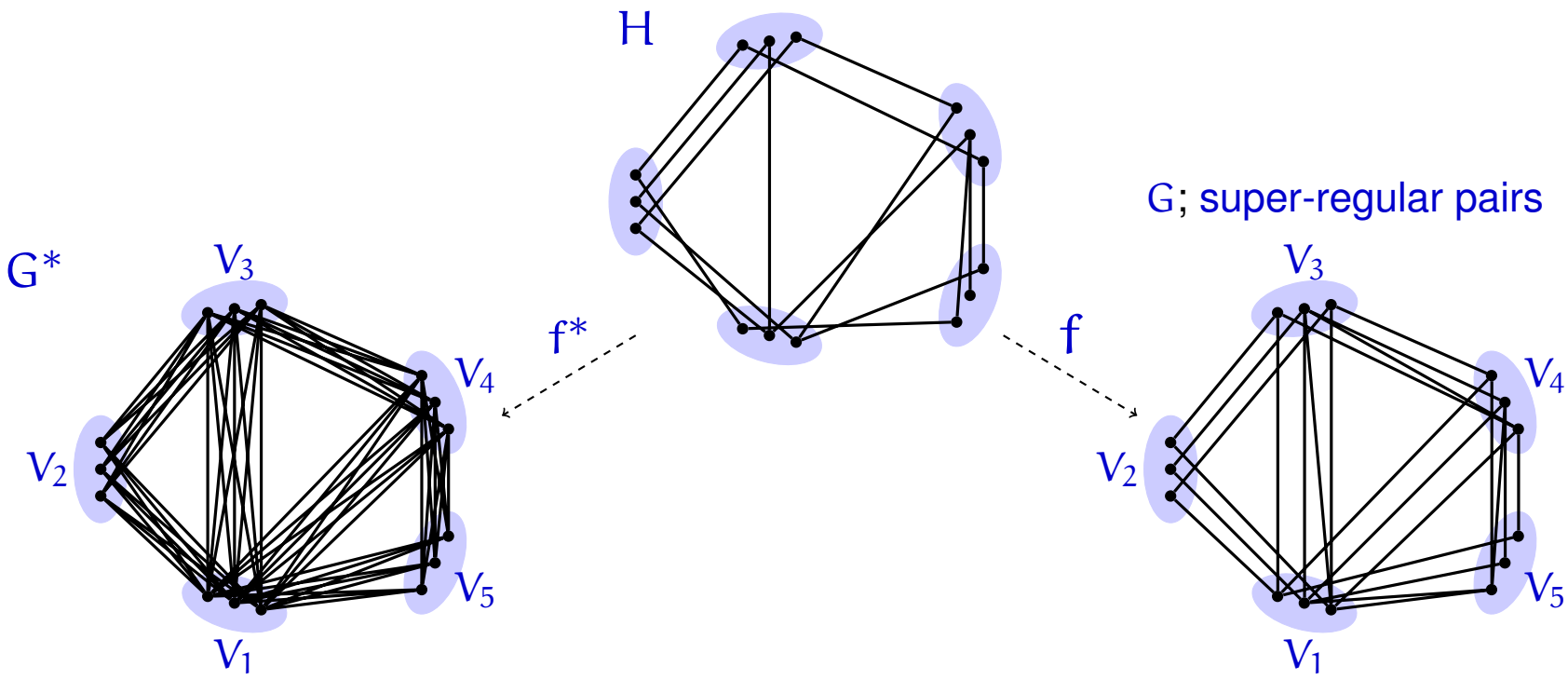
The Blow-up Lemma







The Blow-up Lemma

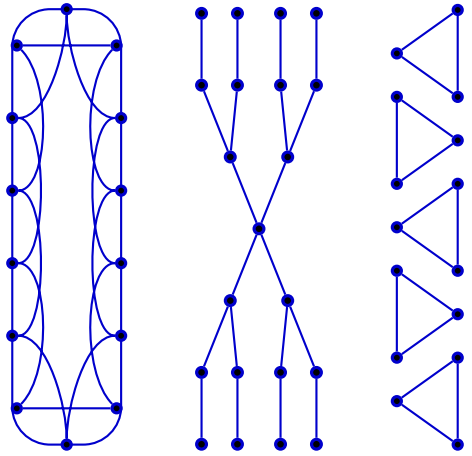


If f^* exists, the f exists.

Applications of the Blow-up Lemma

Do graphs with sufficiently high minimum degree contain . . .

- Pósa & Seymour '74: r-th powers of Hamiltonian cycles?
- Alon & Yuster '96: F-factors (for F fixed)?



Answer: Yes!

- Invented by Komlós, Sárközy & Szemerédi

Spanning subgraphs with constant maximum degree

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The sparse setting

- So far: we have been concerned with dense graphs, i.e., graphs $G = G^n$ with $\Omega(n^2)$ edges.
- *Sparse*: graphs with $o(n^2)$ edges.
- Densities $d(U, U') = e(U, U')/|U||U'|$ in such sparse graphs vanish as $n \to \infty$. Therefore, the regularity condition

  $$\forall U' \subset U,\ W' \subset W \text{ with } |U'|/|U|,\ |W'|/|W| \geq \varepsilon$$
  $$\text{we have } |d(U', W') - d(U, W)| \leq \varepsilon$$

  becomes trivial.
- Easy fix: normalize the density by dividing by $p = p(n) = e(G)\binom{n}{2}^{-1}$.

Szemerédi's regularity lemma, sparse version

Definition 2. $B = (U, W; E)$ *is* $(\varepsilon, d; p)$*-regular if for all* $U' \subset U$ *and* $W' \subset W$ *with* $|U'|/|U|, |W'|/|W| \geq \varepsilon$*, we have* $|d_p(U', W') - d| \leq \varepsilon$*, where*

$$d_p(U', W') = \frac{e(U', W')}{p|U'||W'|}.$$

Typical applications: B is a subgraph of the random graph $G(n, p)$ or of a pseudorandom graph of some density p, with $p \to 0$.

Ambient graph: we shall usually have a *sparse*, *ambient graph* Γ.

Szemerédi's regularity lemma, sparse version

Any graph with no 'dense patches' admits a Szemerédi partition with the above notion of ε-regularity.

Definition 3. *Let us say* $G = (V, E)$ *is locally* (η, b)*-bounded if for all* $U \subset V$ *with* $|U| \geq \eta|V|$*, we have*

$$\#\{\textit{edges within } U\} \leq b|E|\binom{|U|}{2}\binom{|V|}{2}^{-1}.$$

Szemerédi's regularity lemma, sparse version

Theorem 4 (The regularity lemma)**.** *For any* $\varepsilon > 0$, $t_0 \geq 1$, *and* b, *there exist* $\eta > 0$ *and* T_0 *such that any locally* (η, b)*-bounded graph* $G = G^n$ *admits a partition* $V = V_1 \cup \cdots \cup V_t$ *such that*

- $t_0 \leq t \leq T_0$,
- $|V_1| \leq \cdots \leq |V_t| \leq |V_1| + 1$,
- (V_i, V_j) *is* $(\varepsilon; p)$*-regular for at least* $(1-\varepsilon)\binom{t}{2}$ *pairs* $ij \in \binom{t}{2}$, *where*

$$p = e(G)\binom{n}{2}^{-1}.$$

- A structure theorem for locally bounded sparse graphs

Embedding lemma; sparse setting

- Thanks to the work of several researchers, we now know how to use the regularity lemma in the sparse setting.

  Contributors: Balogh, Conlon, Fox, Gerke, Gowers, K., Łuczak, Morris, Rödl, Samotij, Saxton, Schacht, Steger, Thomason, Zhao, among others.

- Typical applications: when $G \subset \Gamma$, and $\Gamma = G(n,p)$ or Γ is a strongly pseudorandom graph.

- More precisely: we know suitable embedding lemmas for graphs $H = H^h$ with $h = O(1)$.

- How about *large* H? E.g., $h = n$?

Sparse blow-up lemmas

- **Main result:** suitable blow-up lemmas exist in the sparse setting.
- Statements are technical: we'll discuss some applications first.

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Sparse blow-up applications

1. Universality results for random and pseudorandom graphs
2. Partition-universality results for random and pseudorandom graphs
3. Resilience and robustness of low-bandwidth graphs
4. Maker-Breaker games

Universality

- $\mathcal{H}$: a set of graphs
- A graph G is $\mathcal{H}$-universal if $H \subset G$ for each $H \in \mathcal{H}$.

A universality result (random graphs)

$\mathcal{H}(n, d, \Delta)$: class of graphs $H = H^n$ with $\Delta(H) = \Delta$ and degeneracy d

Theorem 5. *For every d and $\Delta \in \mathbb{N}$ there is C such that if*

$$p \geq C \left(\frac{\log n}{n}\right)^{1/(2d+1)},$$

then the r.g. $G(n, p)$ is a.a.s. $\mathcal{H}(n, d, \Delta)$-universal.

Theorem 6. *For every d, $\Delta \in \mathbb{N}$ and $\gamma > 0$ there is C such that if*

$$p \geq C \left(\frac{\log n}{n}\right)^{1/2d},$$

then the r.g. $G((1+\gamma)n, p)$ is a.a.s. $\mathcal{H}(n, d, \Delta)$-universal.

Remarks

- Strengthen results of Dellamonica, K., Rödl and Ruciński (2015) and Kim and Lee (2013+), Ferber, Nenadov and Peter (2013+). [For $d = 1$, Montgomery can do $p = \widetilde{O}(1/n)$.]
- One of the forms of the blow-up lemma is tailored to handle graphs with low degeneracy.

A universality result (pseudorandom graphs)

Pseudorandom concept we shall use: G is (p, β)-*bi-jumbled* if, for all U, $W \subset V(G)$, we have

$$\left|e(U, W) - p|U||W|\right| \leq \beta\sqrt{|U||W|}.$$

Thomason (1987): (p, β)-jumbled graphs

Remarks:

(*i*) graph discrepancy results imply that if $G = G^n$ is (p, β)-bi-jumbled, then $\beta \geq \sqrt{np(1-p)}/80$ as long as $p(1-p) \geq 1/n$.

(*ii*) (n, d, λ)-graphs are $(d/n, \lambda)$-bi-jumbled.

(*iii*) Shall just say *jumbled* for *bi-jumbled*.

A universality result (pseudorandom graphs)

$\mathcal{H}(n, \Delta)$: class of graphs $H = H^n$ with $\Delta(H) = \Delta$

Theorem 7. *For every $\Delta \geq 2$ there is $c > 0$ such that, for any $p > 0$, if $\beta \leq cp^{3\Delta/2+1/2}n$, then any n-vertex (p, β)-jumbled graph G with $\delta(G) \geq \frac{1}{2}pn$ is $\mathcal{H}(n, \Delta)$-universal.*

A partition-universality result (random graphs)

G is *r-partition universal for $\mathcal{H}$* if in any r-colouring of $E(G)$ there is a colour class which is $\mathcal{H}$-universal.

$\mathcal{H}(n, d, \Delta)$: class of graphs $H = H^n$ with $\Delta(H) = \Delta$ and degeneracy d

Theorem 8. *For every r, d, and $\Delta \in \mathbb{N}$, there is C such that if*

$$p \geq C \left(\frac{\log n}{n}\right)^{1/2d},$$

then a.a.s. $G(Cn, p)$ is r-partition universal for $\mathcal{H}(n, d, \Delta)$.

- ▷ Strengthens a result of K., Rödl, Schacht and Szemerédi (2011).

A partition-universality result (pseudorandom graphs)

$\mathcal{H}(n, \Delta)$: class of graphs $H = H^n$ with $\Delta(H) = \Delta$

Theorem 9. *For every r and $\Delta \in \mathbb{N}$ there is $c > 0$ such that if $p > 0$ and G is an n/c-vertex graph that is $\left(p, p^{3\Delta/2+1/2}n\right)$-jumbled, then G is r-partition universal for $\mathcal{H}(n, \Delta)$.*

- ▷ Answers a question of K., Rödl, Schacht and Szemerédi (2011).

The bandwidth of a graph

$H = H^n$ has bandwidth $bw(H)$ at most b if there is an ordering $x_1, \ldots, x_n$ of the vertices of H such that every edge $x_i x_j$ of H is such that $|j - i| \leq b$.

Naturally,

$$bw(H) = \min b,$$

with b as above.

The bandwidth theorem

Theorem 10 (Böttcher, Schacht and Taraz (2009))**.** *For every* $\gamma > 0$ *and* Δ *there is* $\beta > 0$ *such that for all sufficiently large* n *the following holds. If* $H = H^n$ *has* $\Delta(H) \leq \Delta$ *and* $\mathrm{bw}(H) \leq \beta n$*, and* $G = G^n$ *is such that*

$$\delta(G) \geq \left(1 - \frac{1}{\chi(H)} + \gamma\right) n,$$

then $H \subset G$*.*

- ▷ Conjectured by Bollobás and Komlós.
- ▷ Generalizes, up to a small error term, several results dealing with spanning graphs in extremal graph theory.

Resilience of low-bandwidth graphs (approximate form)

Theorem 11. *For every $\gamma > 0$ and Δ there is $\beta > 0$ and C such that if*

$$p \geq C\left(\frac{\log n}{n}\right)^{1/\Delta},$$

then a.a.s. $\Gamma = G(n,p)$ has the following property. If H is any $(1-\gamma)n$-vertex graph with $\Delta(H) \leq \Delta$ and $bw(H) \leq \beta n$, and $G = G^n \subset \Gamma$ is such that

$$\delta(G) \geq \left(1 - \frac{1}{\chi(H)} + \gamma\right)pn,$$

then $H \subset G$.

Remarks

- Forthcoming result of Allen, Böttcher, Ehrenmüller and Taraz: the case in which $|V(H)| = n - Cp^{-2}$ if $p \leq 1/\log n$ (which is optimal) and even $H = H^n$ for some H.
- Previous results by Böttcher, K. and Taraz (2013) and Huang, Lee and Sudakov (2012).

Robustness of the Bandwidth Theorem

▷ **Robustness:** proposed by Krivelevich, Lee and Sudakov (2014), who proved that Pósa's result on the Hamiltonicity of $G(n,p)$ with $p = C(\log n)/n$ is very *robust*, in the sense that random subgraphs $G_p = G^n_p$ of *Dirac graphs* are a.a.s. Hamiltonian for $p = C'(\log n)/n$. [Dirac graph: $G = G^n$ with $\delta(G) \geq n/2$].

Robustness of the Bandwidth Theorem

If G is a graph and $0 \le p \le 1$, then G_p is random spanning subgraph of G in which each $e \in E(G)$ is included in G_p with probability p, independently of all other edges.

Theorem 12. *For every $\gamma > 0$ and $\Delta \ge 2$ there are $\beta > 0$ and C such that if $p \ge C\left(\frac{\log n}{n}\right)^{1/\Delta}$, then the following holds. If $H = H^n$ is a graph with $\Delta(H) \le \Delta$ and $bw(H) \le \beta n$, and G is any n-vertex graph with*

$$\delta(G) \ge \left(1 - \frac{1}{\chi(H)} + \gamma\right) n,$$

then a.a.s. $H \subset G_p$.

Make-Breaker games

Maker-Breaker H-game on K^n with bias b: Maker and Breaker take turns to colour the edges of K^n red and blue. In each turn, Maker colours one edge, while Breaker colours b edges (and edges may not be recoloured). Maker's aim: create a red copy of H; Breaker's aim: prevent Maker from doing so.

$\mathcal{H}$-universality game: Maker's graph contains simultaneously each $H \in \mathcal{H}$.

Make-Breaker games

Theorem 13. *For every* $d, \Delta \in \mathbb{N}$ *and* $\delta > 0$ *there is* $c > 0$ *with the following properties.*

(*i*) *If* $b \leq c\left(\frac{n}{\log n}\right)^{1/\Delta}$ *then Maker wins the* $\mathcal{H}(n, \Delta)$*-universality game on* $K^{(1+\delta)n}$ *with bias* b*.*

(*ii*) *If* $b \leq c\left(\frac{n}{\log n}\right)^{1/2d}$ *then Maker wins the* $\mathcal{H}(n, d, \Delta)$*-universality game on* $K^{(1+\delta)n}$ *with bias* b*.*

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An ingredient: inheritance of density and regularity

- Immediate inheritance
- Inheritance by very small sets

Immediate inheritance; dense case

- Suppose have an ε-regular triple $G = (X, U, W; E)$, with all pairs with density d.

Fact 14. *Suppose $d \gg \varepsilon$. Then typical vertices $x \in X$ are such that*

$$(N_G(x) \cap U, N_G(x) \cap W) \tag{1}$$

has density $d(U, W) \pm \varepsilon$. Moreover, the pair is, say, $2\varepsilon/d$-regular.

In particular, (1) induces an edge, and hence $K^3 \subset G$. Even have many such edges, and hence many such K^3. The regularity of pairs of the form (1) is useful in inductive arguments.

Inheritance

Question 15. *Suppose* $B = (U, W; E)$ *is an* ε*-regular pair of density* $d = |E|/|U||W|$*. Suppose* $W' \subset W$ *and* $U' \subset U$*. What can we say about the density and the regularity of* $B[U', W']$*?*

- **Inheritance:** positive results in this direction.
- Immediate if can take

$$\varepsilon \ll |U'|/|U|,\ |W'|/|W|. \tag{2}$$

- In several applications, can take (2).
- What if $|U'| \ll \varepsilon|U|$ or $|W'| \ll \varepsilon|W|$?

▷ Often, U' and W' are neighbourhoods (or intersections of neighbourhoods). If the graphs are **sparse**, then certainly (2) fails.

Inheritance results

Even in the case $|U'|/|U|, |W'|/|W| \ll \varepsilon$,

- ▷ there are inheritance results for subgraphs of r.gs: $B = (U, W; E) \subset \Gamma = G(n, p)$, and
- ▷ there are inheritance results for subgraphs of (p, β)-jumbled graphs Γ: $B = (U, W; E) \subset \Gamma$,

as long as one considers U' and W' that are neighbourhoods or joint neighbourhoods of vertices in Γ.

Random case

- **Dense/classical case:** overwhelming majority of pairs (U', W') induce ε'-regular bipartite graphs $B[U', W']$, with $\varepsilon' \to 0$ as $\varepsilon \to 0$, as long as $|U'|, |W'| \geq C/d$, where $C = C(\varepsilon')$.
- Statement above is true even if $d \to 0$ (as long as we replace ε'-regularity by ε'-lower-regularity) [Gerke, K., Rödl, Steger 2007]

Definition 16. $B = (U, W; E)$ *is* $(\varepsilon, d; p)$*-lower-regular if for all* $U' \subset U$ *and* $W' \subset W$ *with* $|U'|/|U|, |W'|/|W| \geq \varepsilon$*, we have*

$$d_p(U', W') = \frac{e(U', W')}{p|U'||W'|} \geq d - \varepsilon.$$

Inheritance for subgraphs of $G(n,p)$ **[SKIP]**

Set-up 17. *$J = J^{3m}$ is a graph with vertex classes X, U, and W, all of cardinality m; the graphs $J[X,U]$, $J[X,W]$, and $J[U,W]$ are $(\varepsilon, d; p)$-lower-regular, and $J = J[X,U] \cup J[X,W] \cup J[U,W]$.*

Definition 18. *J is ε'-good if $\exists X' \subset X$ with $|X'| \geq (1-\varepsilon')m$ such that $\forall x' \in X'$ we have that $J[N(x') \cap U, N(x') \cap W]$ is $(\varepsilon', d; p)$-regular.*

Inheritance for subgraphs of $G(n,p)$

Theorem 19 (Gerke, K., Rödl, Steger 2007)**.** *For all ε' and $d > 0$, there is $\varepsilon > 0$ such that if $p^2 m \gg 1$ and $pm \gg \log n$, then*

$$\mathbb{P}(\exists J^{3m} \subset G(n,p) \ \varepsilon'\text{-bad}) = o(1).$$

- ▷ Similar statement holds for the setting in which we are concerned with $J^{4m} = J[X,U] \cup J[U,W] \cup J[W,Y]$ and pairs $(x,y) \in X \times Y$.

Inheritance for subgraphs of (p, β)-jumbled graphs

(*i*) Recall Set-up 17.

(*ii*) Suppose $J = J^{3m} \subset \Gamma$, with Γ $(p, \gamma p^d n)$-jumbled, for some $d \geq 1$.

(*iii*) Furthermore, suppose $m = \Omega(p^f n)$ and $d \geq f + 3$.

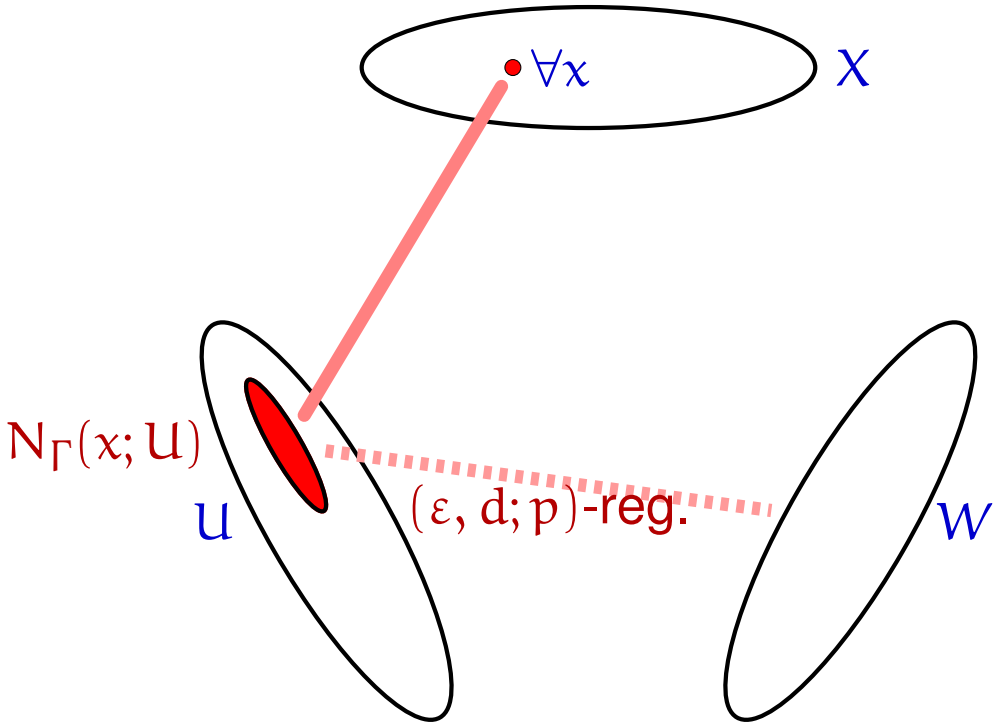
(*iv*) (Recall) J is ε'-good if $\exists X' \subset X$ with $|X'| \geq (1-\varepsilon')m$ such that $\forall x' \in X'$ we have that $J[N(x') \cap U, N(x') \cap W]$ is $(\varepsilon', d; p)$-regular.

Theorem 20 (K., Rödl, Schacht & Skokan (2010)). *For all ε' and $d > 0$, there are ε and $\gamma > 0$ such that if (ii) and (iii) hold, then J is ε'-good.*

One-sided inheritance in Γ **[RESUME]**

Definition 21. *Let* (X, U, W) *be a triple of pairwise disjoint vertex sets in* $G \subset \Gamma$*. We say that* (X, U, W) *has one-sided* $(\varepsilon, d; p)$*-inheritance if for each* $x \in X$ *the pair* $(N_\Gamma(x; U), W)$ *is* $(\varepsilon, d; p)$*-regular in* G*.*

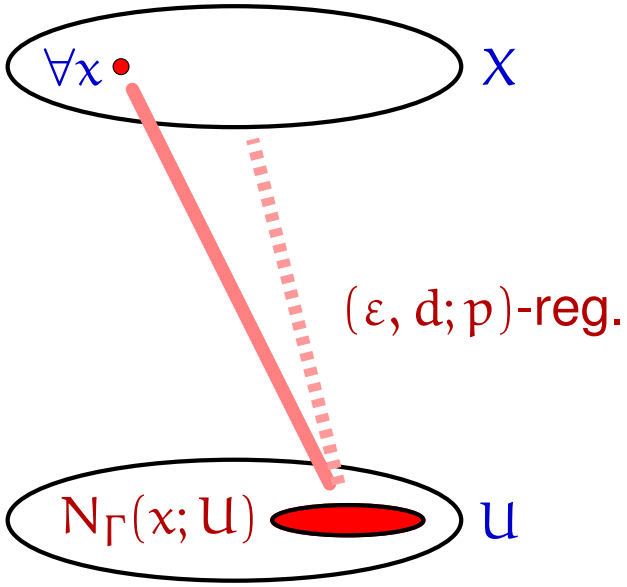
One-sided inheritance in Γ



One-sided inheritance in Γ

Definition 22. *Let* (X, U) *be a pair of vertex sets in* $G \subset \Gamma$*, with* X *and* U *disjoint. We say that* (X, U) *has one-sided* $(\varepsilon, d; p)$*-inheritance if for each* $x \in X$ *the pair* $(N_\Gamma(x; U), X)$ *is* $(\varepsilon, d; p)$*-regular in* G*.*

One-sided inheritance in Γ

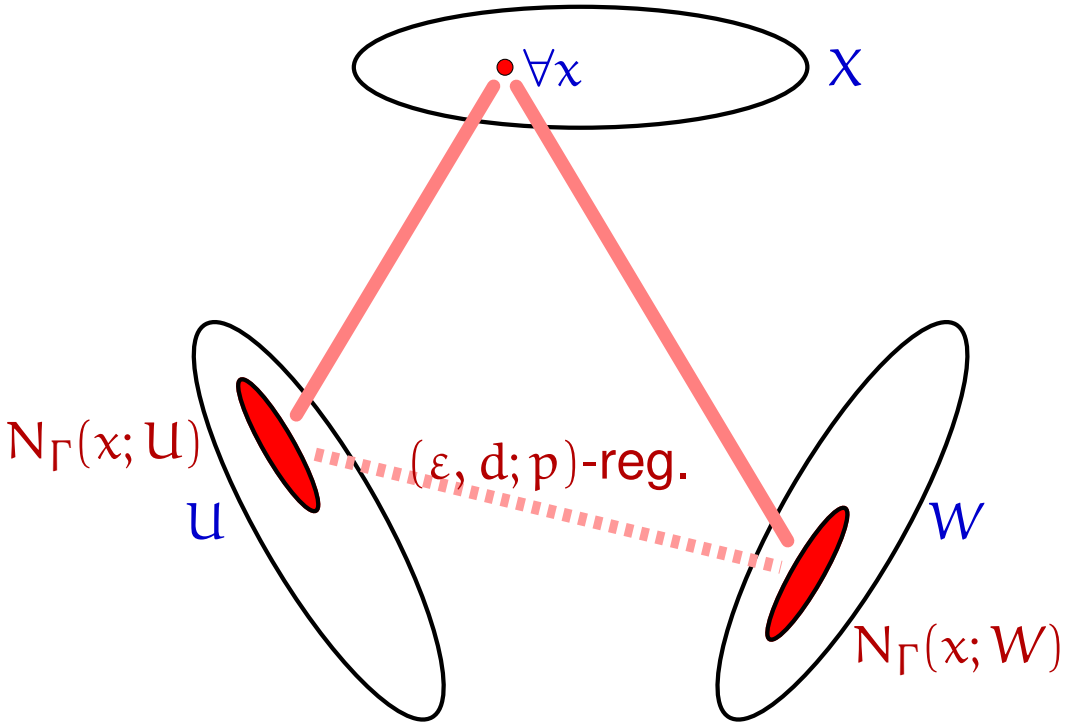


▷ To talk about one-sided inheritance for (X, U, W) and (X, U) at the same time, we allow the triple (X, U, X).

Two-sided inheritance in Γ

Definition 23. *Let (X, U, W) be a triple of pairwise disjoint vertex sets in $G \subset \Gamma$. We say that (X, U, W) has two-sided $(\varepsilon, d; p)$-inheritance if for each $x \in X$ the pair $(N_\Gamma(x; U), N_\Gamma(x; W))$ is $(\varepsilon, d; p)$-regular in G.*

Two-sided inheritance in Γ



Outline of talk

1. (Lecture I & II) The regularity method; the blow-up lemma
2. (Lecture III) The regularity method in the sparse setting
3. Applications of the blow-up lemmas in the sparse setting
4. Inheritance of regularity
5. Statement of the sparse blow-up lemmas

Sparse super-regularity

Recall: $B = (U, W; E)$ is $(\varepsilon, d; p)$-lower-regular if for all $U' \subset U$ and $W' \subset W$ with $|U'|/|U|, |W'|/|W| \geq \varepsilon$, we have $d_p(U', W') \geq d - \varepsilon$, where $d_p(U', W') = e(U', W')/p|U'||W'|$.

Definition 24 (sparse super-regularity in random graphs)**.** *A pair (U, W) in $G \subset \Gamma$ is called $(\varepsilon, d; p)$-super-regular in G if it is $(\varepsilon, d; p)$-lower-regular and, for every $u \in U$ and $w \in W$, we have*

$$\deg_G(u; W) > (d - \varepsilon) \max\{p|W|, \deg_\Gamma(u; W)/2\}.$$
$$\deg_G(w; U) > (d - \varepsilon) \max\{p|U|, \deg_\Gamma(w; U)/2\}.$$

A simplified blow-up lemma for random graphs

The objects:

- $\Gamma = G(n,p)$, where $p \geq C(\log n/n)^{1/\Delta}$,
- $G = G^n \subset \Gamma$ and $H = H^n$ with $\Delta(H) \leq \Delta$,
- H is r-partite, with r-partition $V(H) = X_1 \cup \cdots \cup X_r$. Moreover, $V(G) = V_1 \cup \cdots \cup V_r$ is an equitable partition with $|V_i| \geq |X_i|$ for all $i \in [r]$.

A simplified blow-up lemma for random graphs

Furthermore, we suppose that the following conditions hold:

(*i*) (V_i, V_j) is $(\varepsilon, d; p)$-super-regular in G for each $i, j \in [r]$ with $i \neq j$.

(*ii*) (V_i, V_j, V_k) has one-sided $(\varepsilon, d; p)$-inheritance for each $i, j, k \in [r]$ with $i \neq j$ and $j \neq k$.

(*iii*) (V_i, V_j, V_k) has two-sided $(\varepsilon, d; p)$-inheritance for each $i, j, k \in [r]$ with i, j and k all distinct.

Conclusion: Then H is a subgraph of G.

Theorem 25. *For all $\Delta, r \in \mathbb{N}$ and $d > 0$ there exist $\varepsilon > 0$ and C such that if $p \geq C(\log n/n)^{1/\Delta}$, then the random graph $\Gamma = G(n, p)$ asymptotically almost surely satisfies the above.*

Full blow-up lemma for random graphs

Features include:

(*i*) Partition of $V(G)$ is allowed to be just approximately balanced.

(*ii*) Have two 'reduced graphs' R and R' to encode regular and super-regular pairs ($R' \subset R$).

(*iii*) Require two-sided regularity inheritance only where triangles of H need to be embedded.

(*iv*) The required regularity, i.e., the value of ε, does not depend on r (number of parts in the partitions of H and G), but only on $\Delta(R')$. (New for $p = 1$ also)

(*v*) Image restrictions are permitted.

Full blow-up lemma for random graphs, cont.

We have a specific version for embedding *degenerate* graphs H:

- ▷ E.g., for embedding $H = H^n$ with $\Delta(H) = \Delta$ and degeneracy D into suitably compatible $G = G^n \subset G(n,p)$ with

$$p \geq C_\Delta \left(\frac{\log n}{n}\right)^{2D+1}.$$

Full blow-up lemma for pseudorandom graphs

Rough statement:

- ▷ We are able to embed $H = H^n$ with $\Delta(H) = \Delta$ into suitably compatible $G^n \subset \Gamma$, when $\Gamma = \Gamma^n$ is a $(p, o(p^{(3\Delta+1)/2}n))$-jumbled graph.